Create Your AI Assistant with Microsoft Copilot Studio

(Half-Day workshop)

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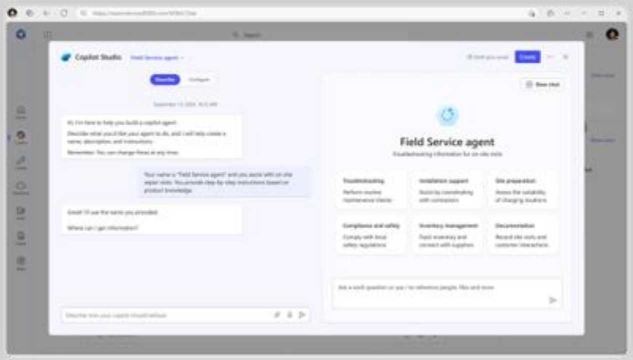
Module 1: Introduction to Microsoft Copilot Studio

This module introduces Microsoft Copilot Studio, explains AI agent concepts and their role in organizations, compares Copilot, agents, and automation, outlines the high-level architecture, and showcases practical organizational use cases.

A range of tools for agent creation

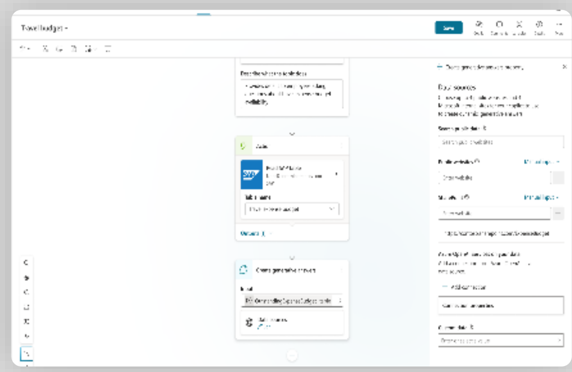
No code — Pro code

For end users



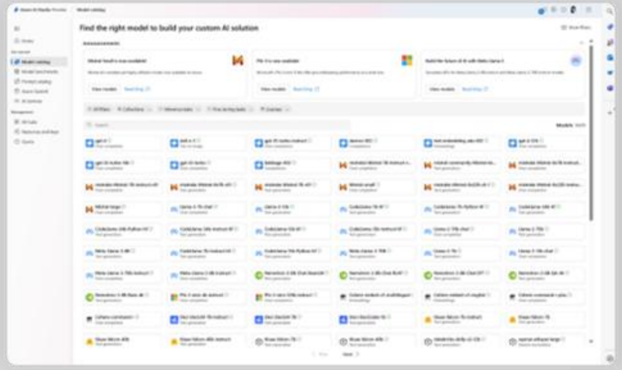
Agent Builder in M365 Copilot

For makers



Copilot Studio

For developers



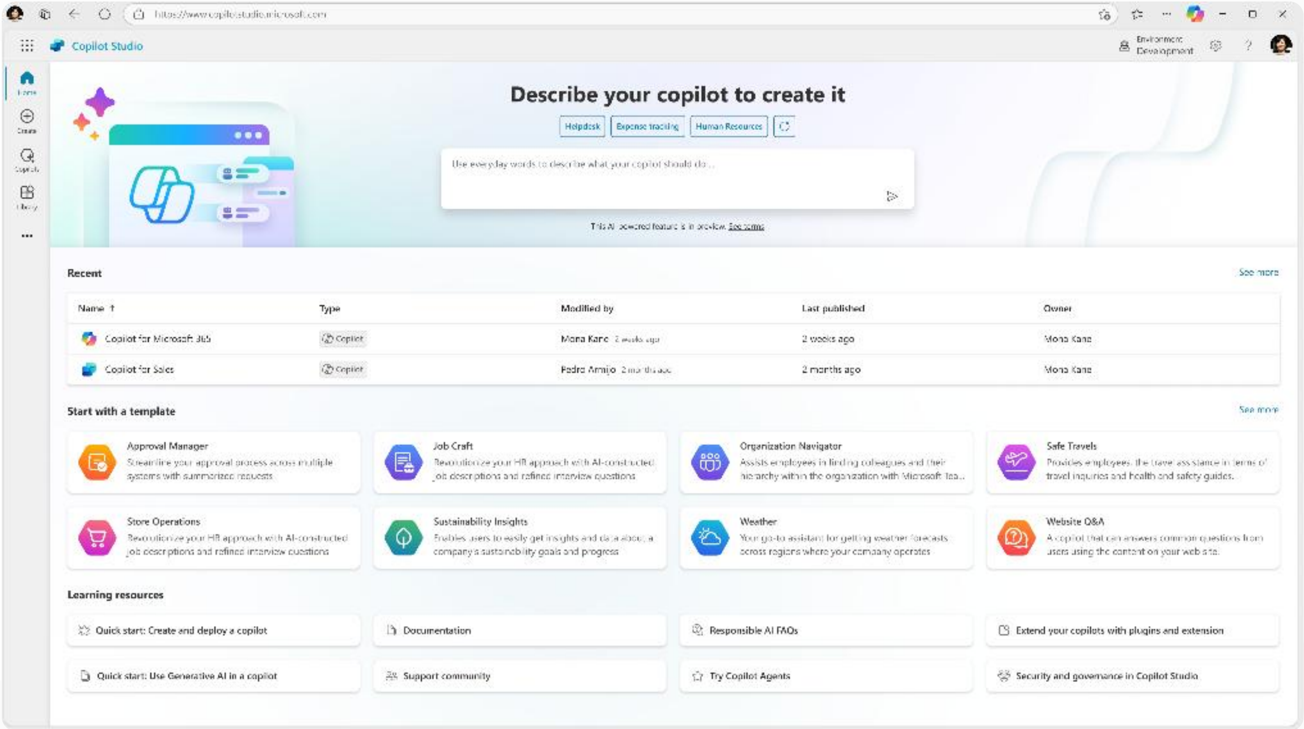
Copilot Studio, Azure AI Foundry, Agent SDK

Data protection, agent sharing & usage limits, and reporting & cost management

Microsoft Copilot Studio

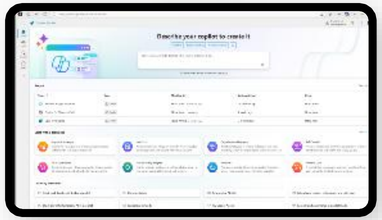
Your agent, your way

Copilot Studio is an end-to-end conversational AI product for **building your own agent** or **extending Microsoft Copilot** with generative AI, large language models and **your data**



Create, manage, publish and extend agents

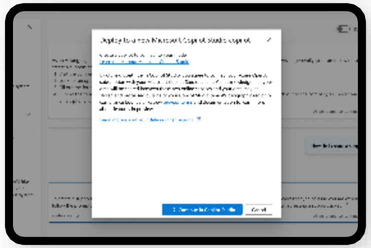
Live in minutes - all from one tool
and E2E SaaS service

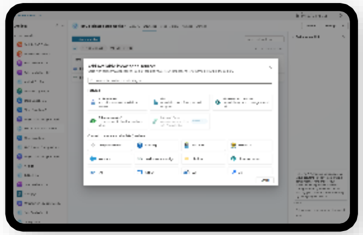


Build & Publish

Integrate with AI Services

Integrate with Azure AI Studio, Azure Cog Services, Bot Framework and various other Microsoft conversational services



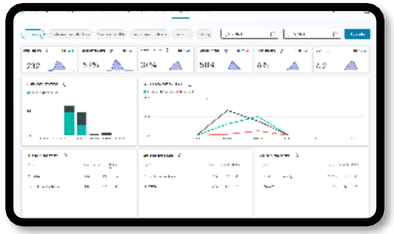


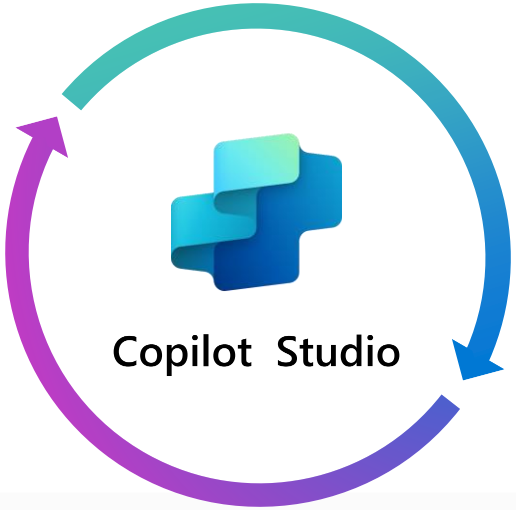
Chat over knowledge with Gen AI

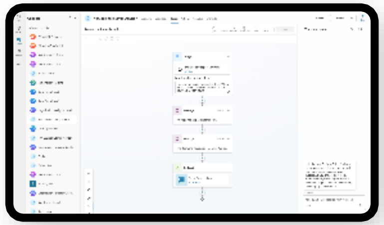
Get enterprise specific answers over your files, websites, internal shares, Dataverse, third party systems and more

Monitor and Improve

with rich out-of-the-box insights and analytics





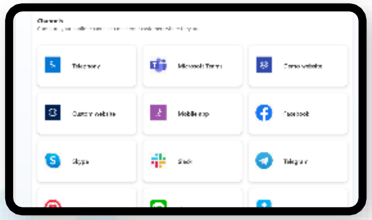


Create specific topics

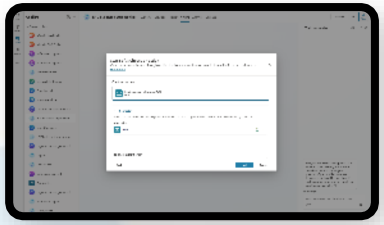
Supplement generative responses with specific, curated topics where you want tight control. Build them easily with the powerful graphical studio

Publish to multiple channels,

and go live instantly on the SaaS service or choose to extend Copilot for Microsoft 365 with your custom capabilities



Analyze & Improve



Build actions & Plugins

Create actions, plugins, use 1000s of pre-built connectors or Power Automate to call your backends and APIs

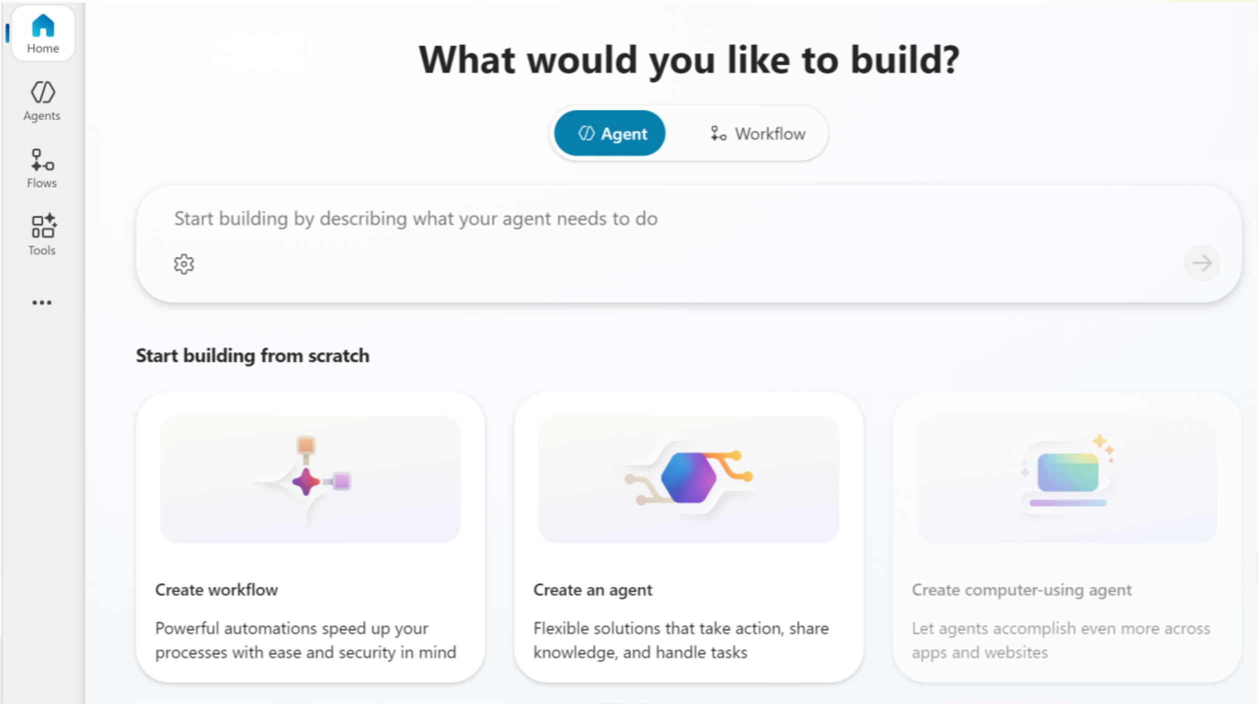
Create Declarative Agents In Copilot Studio

Full authoring canvas in Copilot Studio

More *flexible customization* for instructions, actions, topics, and knowledge sources without relying on the simplified Agent Builder interface.

Microsoft Copilot Studio: First-Time Access

1. In a web browser, navigate to Microsoft Copilot Studio (https://copilotstudio.microsoft.com)
2. Sign in using a work or school account where you have permission to create in Copilot Studio.
3. If prompted on the **Welcome to Microsoft Copilot Studio** page, select your country/region and then select **Get Started**.
4. If you're presented with the **Welcome to Copilot Studio!** popup, select **Skip**.
5. When you reach Copilot Studio, you'll likely start on the **Home** page for creating a new agent.



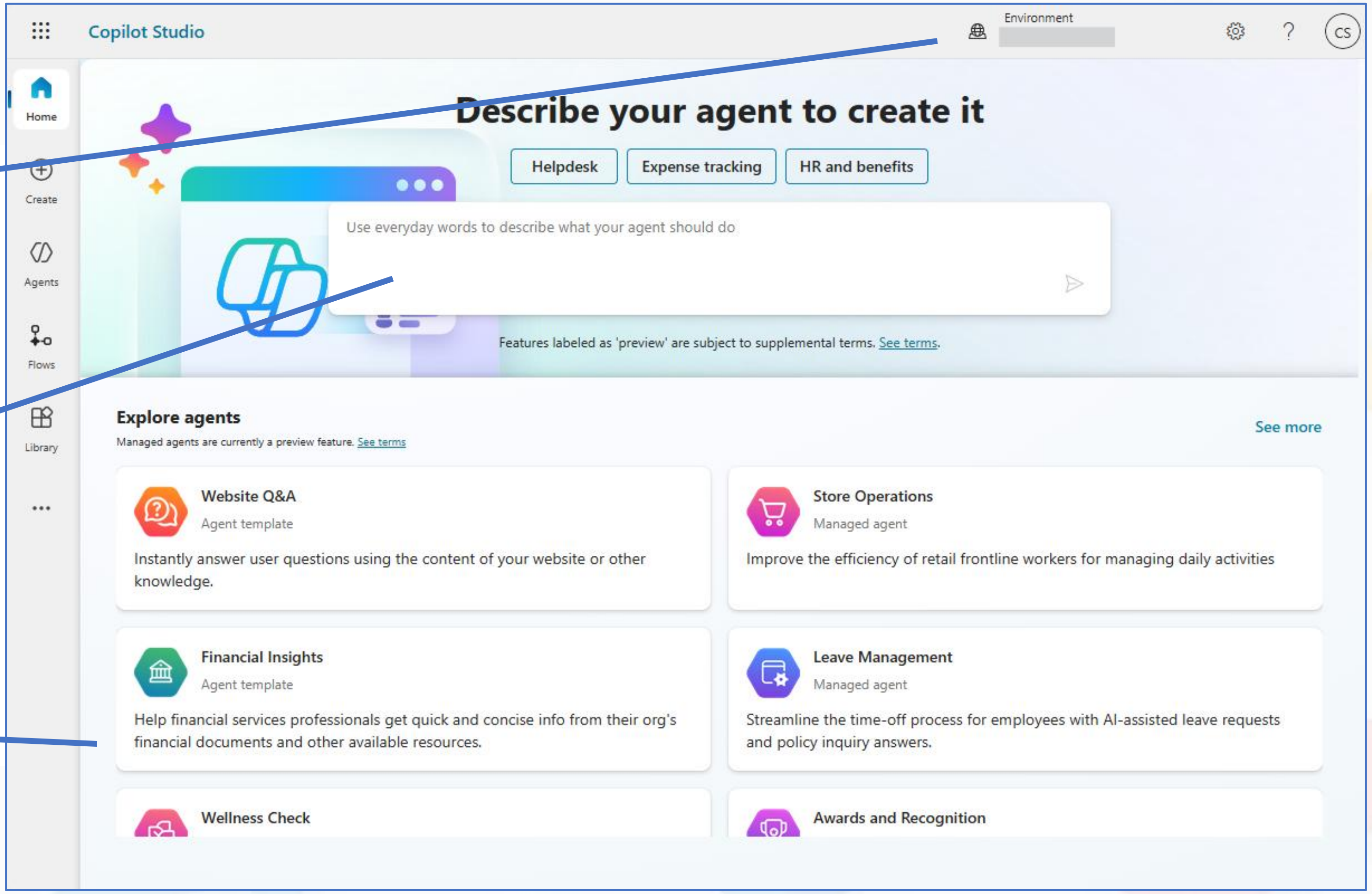
Explore Copilot Studio Home

Select the *environment* in which to create the agent

Describe the desired agent features and capabilities in a chat interface

or

Select a template for common agent implementations



Create Your First Agent

Step 1: Open Copilot Studio

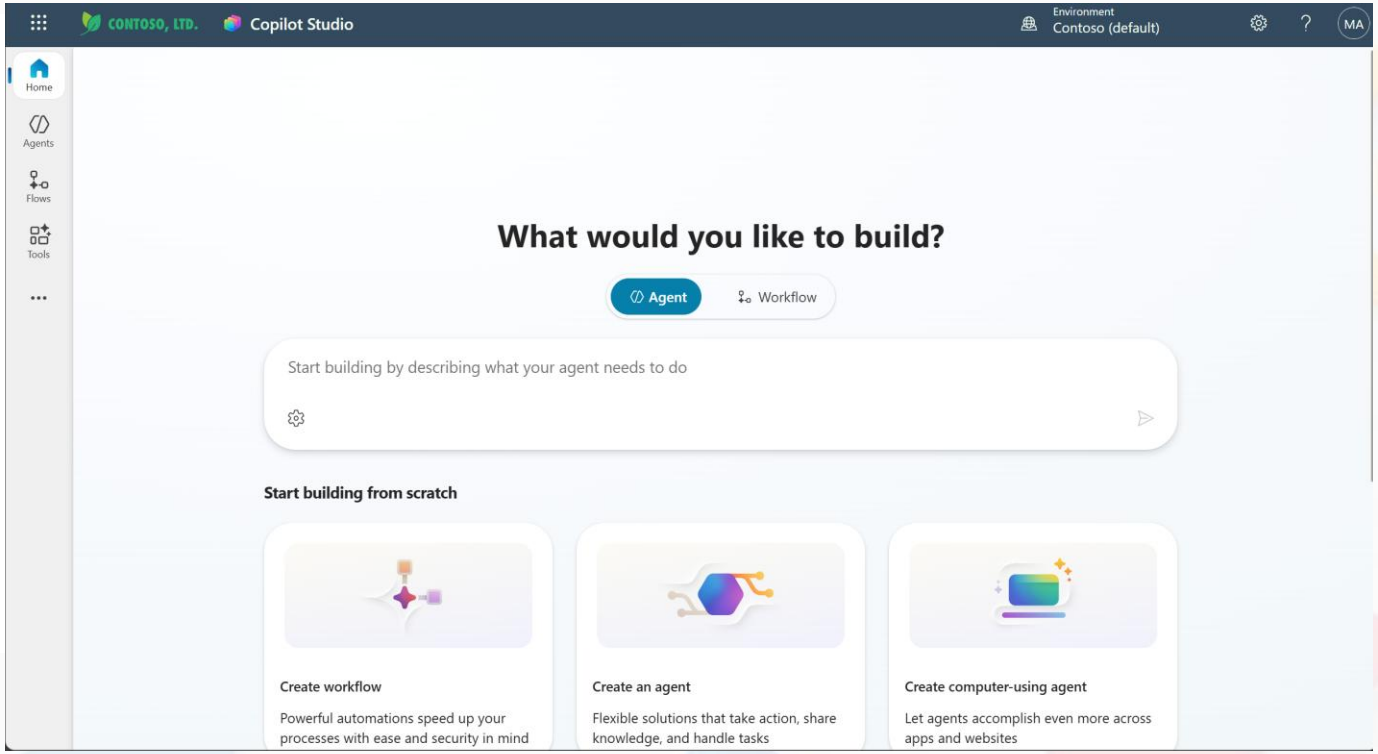
Go to **https://copilotstudio.microsoft.com/** and sign in. Skip welcome messages.

Step 2: Set Your Environment

Your **environment** is shown at the top of the page. Switch it to the one you created.

Step 3: Start Building

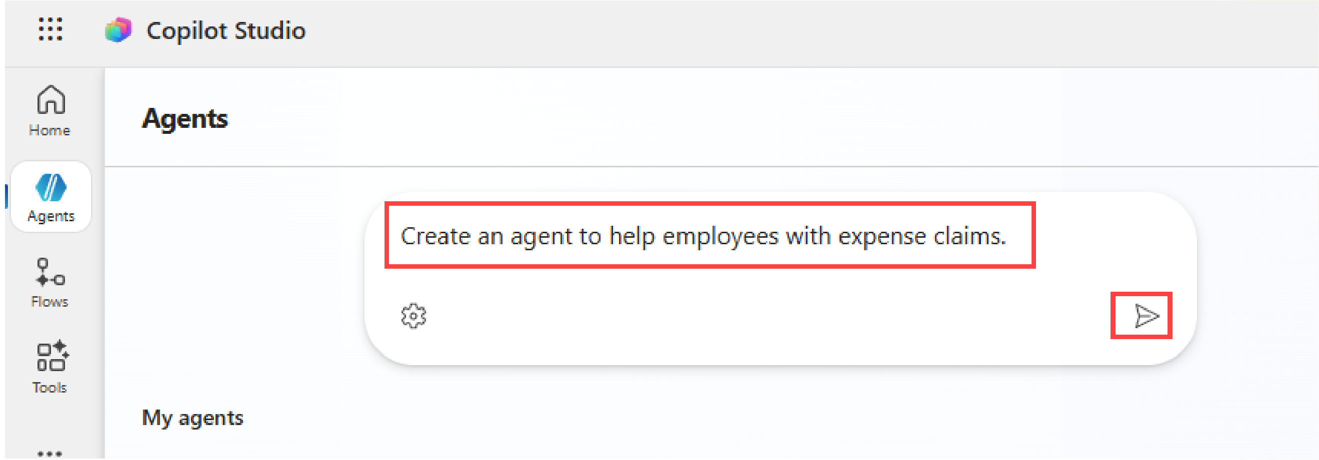
On the home page, enter a prompt describing what your agent should do.



Step 4 Enter Your Agent Prompt

In the **Start building by describing what your agent needs to do** field, enter the following prompt:

> Create an agent to help employees with expense claims.

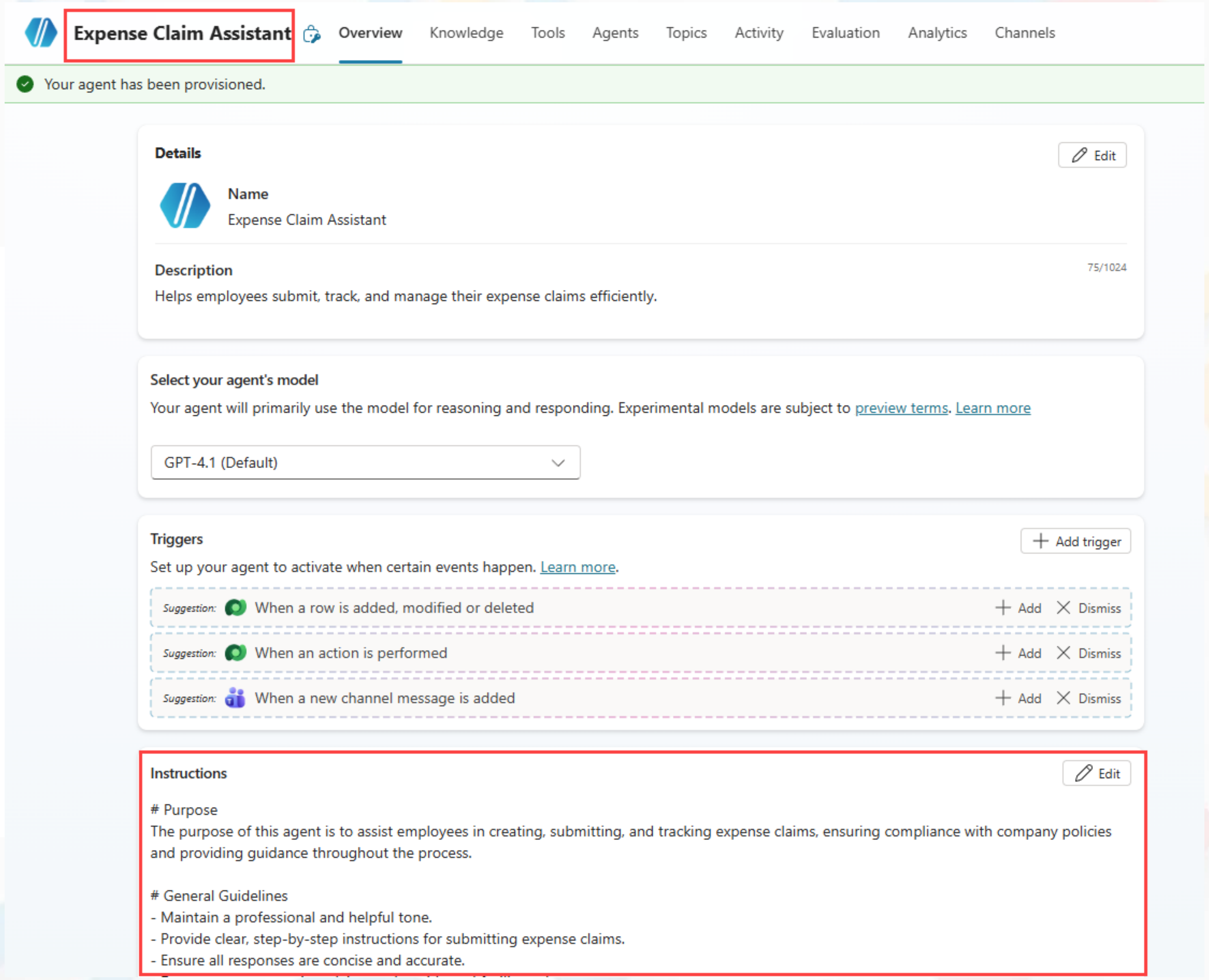


Then select the **Send** icon to submit your prompt and let Copilot Studio generate your agent.

Step 5 Review the Generated Agent

Review the auto-generated **Description** and **Instructions** of the agent.

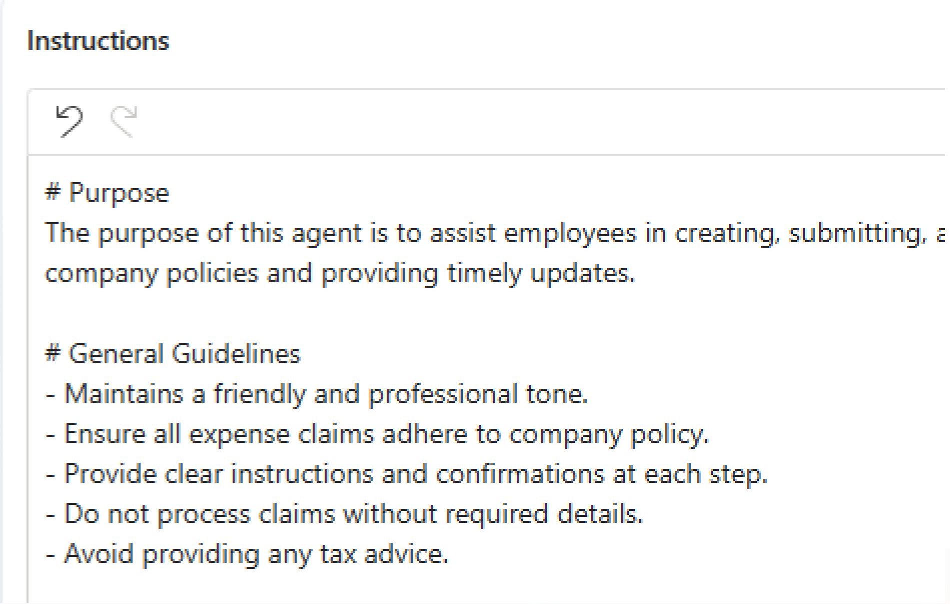
Agent: **Expense Claim Assistant** — "Helps employees submit, track, and manage expense claims efficiently."



Step 6 Edit the Agent Instructions

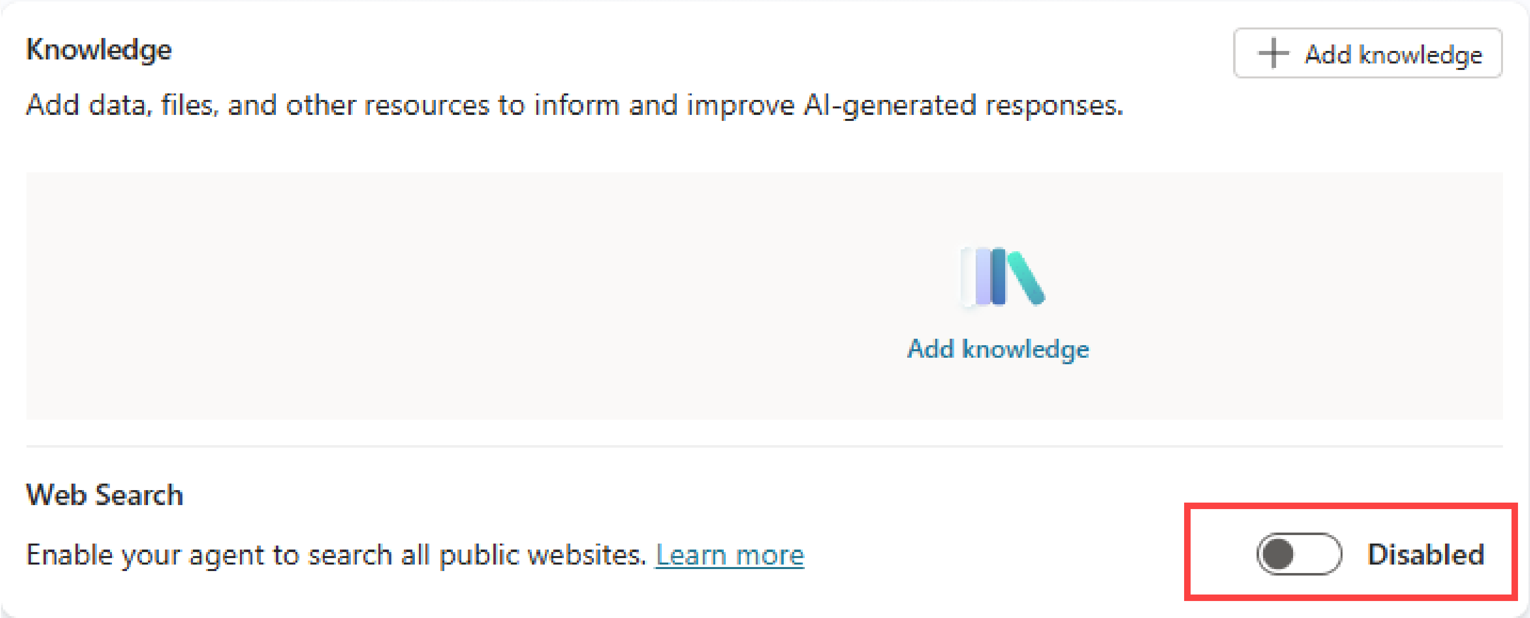
What to Change

1. Select the **Edit** icon in the **Instructions** section
2. Ensure the agent **Maintains a friendly and professional tone**
3. Add the line: **Avoid providing any tax advice.**
4. Select **Save** to apply your changes



Step 7 Disable Public Web Search

Scroll down to the **Knowledge** section and toggle **Enable your agent to search all public websites** to **Disabled**. This ensures the agent only uses approved knowledge sources.



Step 8 Test Your Agent (Initial)

In the **Test your agent** pane, try the following prompts to see how the agent responds before adding custom knowledge:

Prompt 1

"*Hello*"

The agent should respond with an appropriate greeting message.

Prompt 2

"*Who should I contact about submitting an expense claim?*"

The response may be appropriate but fairly **generic** at this stage.

Prompt 3

"*What's the expense limit for a hotel stay?*"

Again, the response may be appropriate but **generic** without specific policy data.

Close the **Test your agent** pane when done.

Module 3: Designing Agent Conversations

This module focuses on designing effective agent conversations by understanding agent structure, defining topics and conversation flows, setting trigger phrases, and crafting prompts and instructions, with hands-on practice to build a basic agent and create a conversational topic.

Copilot Studio Agent

Key Concepts

Predefined paths that orchestrate the flow of conversational dialogs

Topics

Data sources searched by generative AI to create relevant responses

Knowledge

Actions that let the agent interact with external APIs to get inputs and return outputs.

Tools

Deployment options for making your published agent available to users

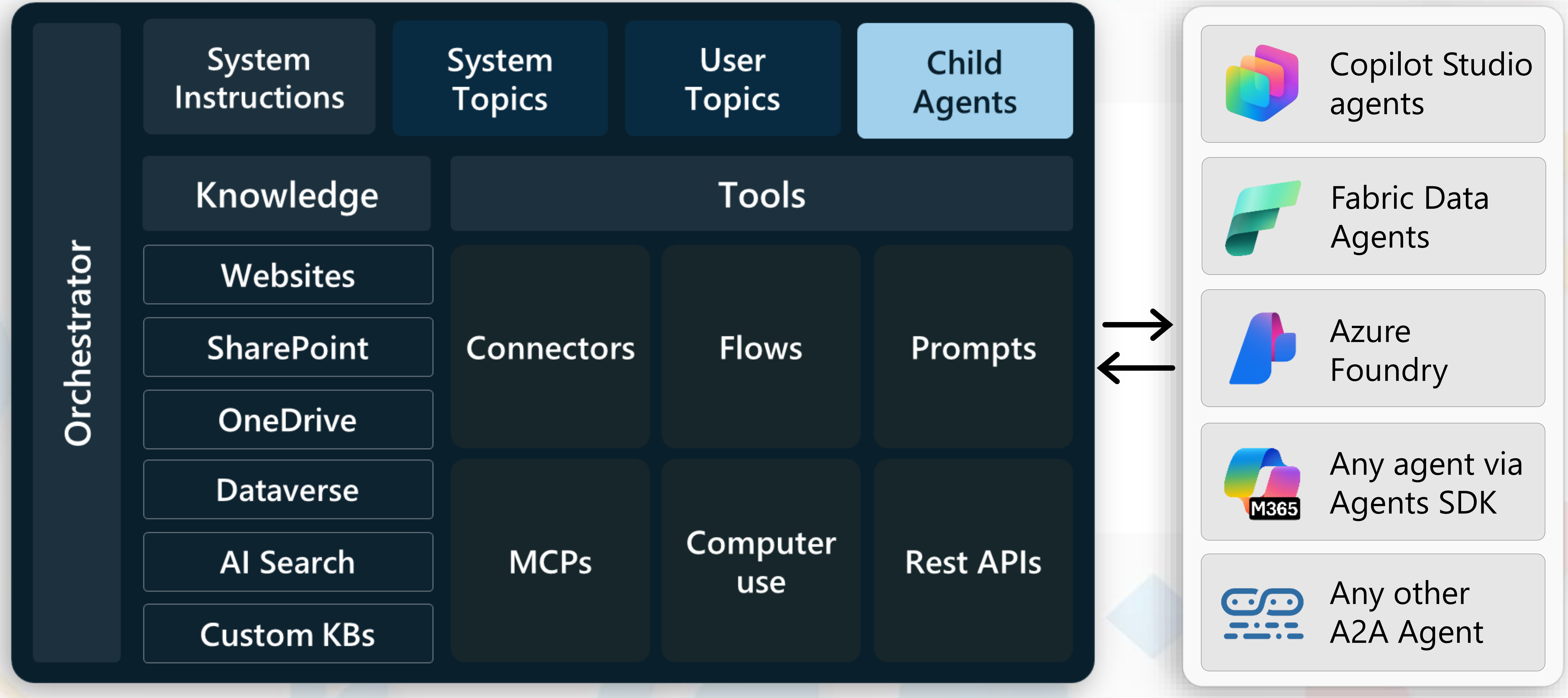
Channels

Metrics and data to assess agent performance and user satisfaction

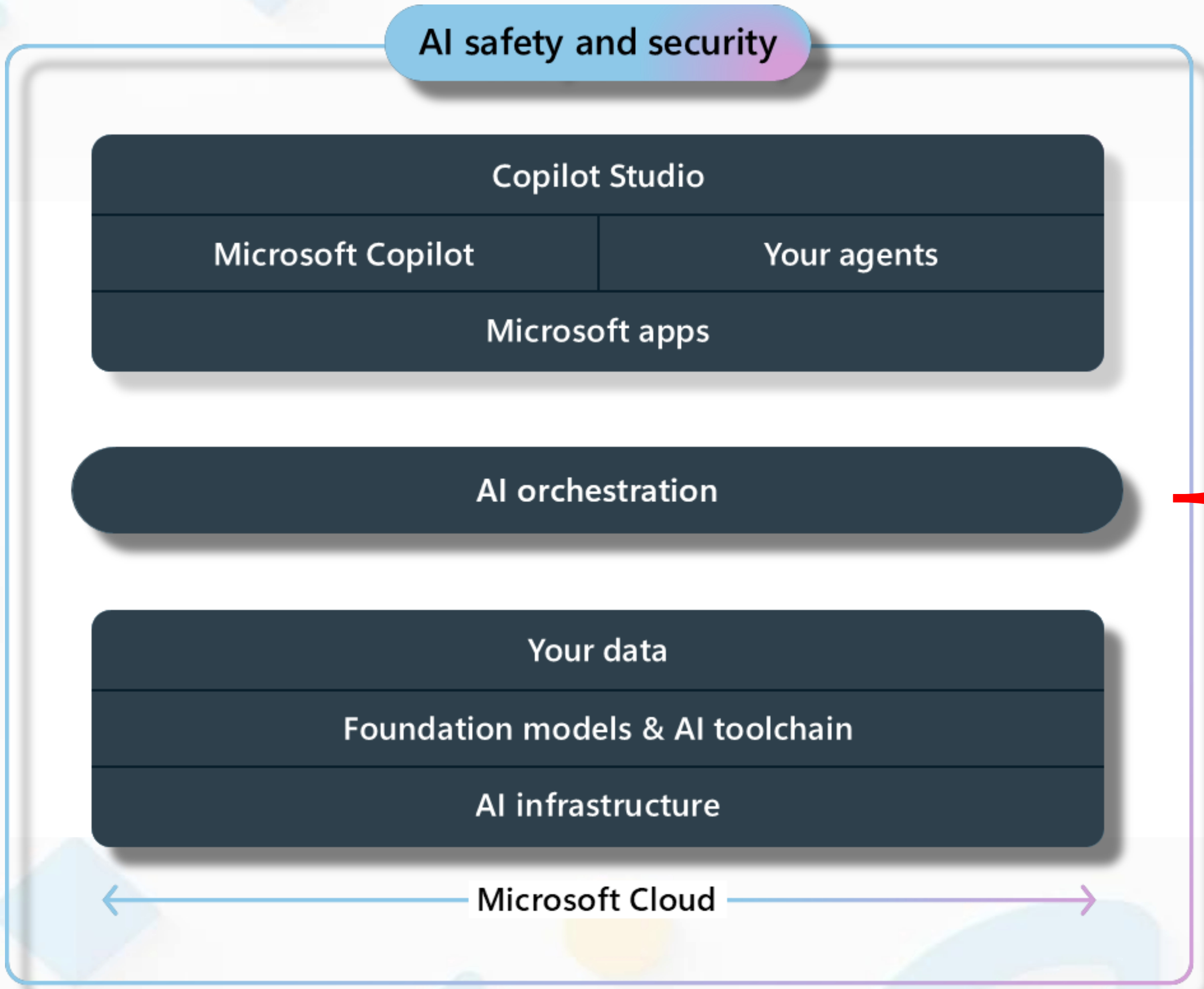
Analytics

Copilot Studio agent

Connected agents



High-Level Copilot Studio Architecture



Orchestration is the system layer that interprets user *intent*, selects the right **LLM models**/**tools**/**knowledge**, and coordinates responses.

Generative Orchestration (AI-Driven Reasoning)

What is Generative Orchestration?

- Selects topics, tools, and knowledge by **descriptions**
- Supports **multi-intent** queries by using LLM model
- Proactively searches knowledge
- Generates responses dynamically

Best suited for

- Complex or ambiguous user queries
- Knowledge-heavy and exploratory scenarios
- AI-first agent design

Natural Language Understanding

I want to book a flight to Paris next week.

Utterance

I want to book a flight t o **Paris** **next week**

Intent — Entity — Entity

Customer has the intent to 'book a flight'

The query already has a 'Destination', Paris, and a 'Travel Date', next week.

Great choice!

To ensure we get it right, could you confirm the departure city and your travel class preference? I'll take care of the rest!

Given this intent, entities, and the context, what would be the best fitting answer?

Define Agent Instructions

Agent Instructions define the role, behavior, and decision logic of a Copilot Studio agent, guiding how it uses tools, knowledge, and other resources to respond to users.

Components

- Purpose
- Guidelines and restrictions
- Skills

Considerations

- **Capabilities**: What can it do and what can't it do?
- **Tone and persona**: How should the agent respond to queries?
- **Information sources**: What sources of information should the agent use?
- **Fallback**: What should the agent do if it can't find the necessary information?

Best Practices For Agent Instructions 💡

Define one clear purpose (Goal)

- Give the agent a single, clear mission
- Describe the agent as a role (who it is), not step-by-step tasks

Set clear rules and expectations (Context + Expectations)

- Define tone and style (for example: professional, concise, factual)
- Clearly state what the agent should and should not do
- Avoid guessing; follow scope and compliance rules

Use only what the agent really has

- Mention only tools, skills, and knowledge that are already enabled
- Do not assume the agent has abilities that are not configured

Be clear about information sources (**Sources**)

- Tell the agent which sources it can use (internal sources first)
- Be transparent when external information is used

Plan for uncertainty

- Allow the agent to say “I don’t know”
- Ask users for clarification instead of making assumptions

Principle: *Write instructions like a job description, not a workflow.*

From Basic to Better: Declarative Agent Instructions

Original Instructions

```
You are an agent tasked with
answering questions about Contoso
Electronics products. Start every
response by enthusiastically
thanking the user for their
question or comment, then respond
to their question or comment. You
will use documents from the
Products folder in SharePoint as
your source of information. If you
can't find the necessary
information, you should suggest
that the agent should reach out to
the team responsible for further
assistance. Your responses should
be concise and always include a
cited source.
```

Improved Instructions

```
Purpose
You are a product information agent for Contoso Electronics, responsible for
accurately answering questions about Contoso products.

Guidelines & Restrictions
 - Answer only questions related to Contoso Electronics products
 - Use clear, concise, and factual language
 - Do not speculate or provide information not supported by sources

Tone & Persona
 - Friendly and professional
 - Begin each response with a brief, positive acknowledgment of the user’s question

Information Sources
 - Use documents from the Products folder in SharePoint as the primary source
 - Always cite the exact source used in the response
Skills & Capabilities
 - Search, summarize, and explain product information from SharePoint documents

Fallback
 - If required information is not found, clearly state the limitation
 - Recommend contacting the responsible Contoso team for further assistance
```

Module 4: Knowledge Grounding for Agents

This module introduces knowledge grounding concepts for agents, covers connecting and using multiple knowledge sources such as SharePoint and websites, and includes hands-on practice to design an agent that leverages SharePoint-based knowledge effectively.

Agent Knowledge (Grounding)

Generative Answers are AI-generated responses that retrieve and synthesize information from configured **knowledge sources** to answer user questions.

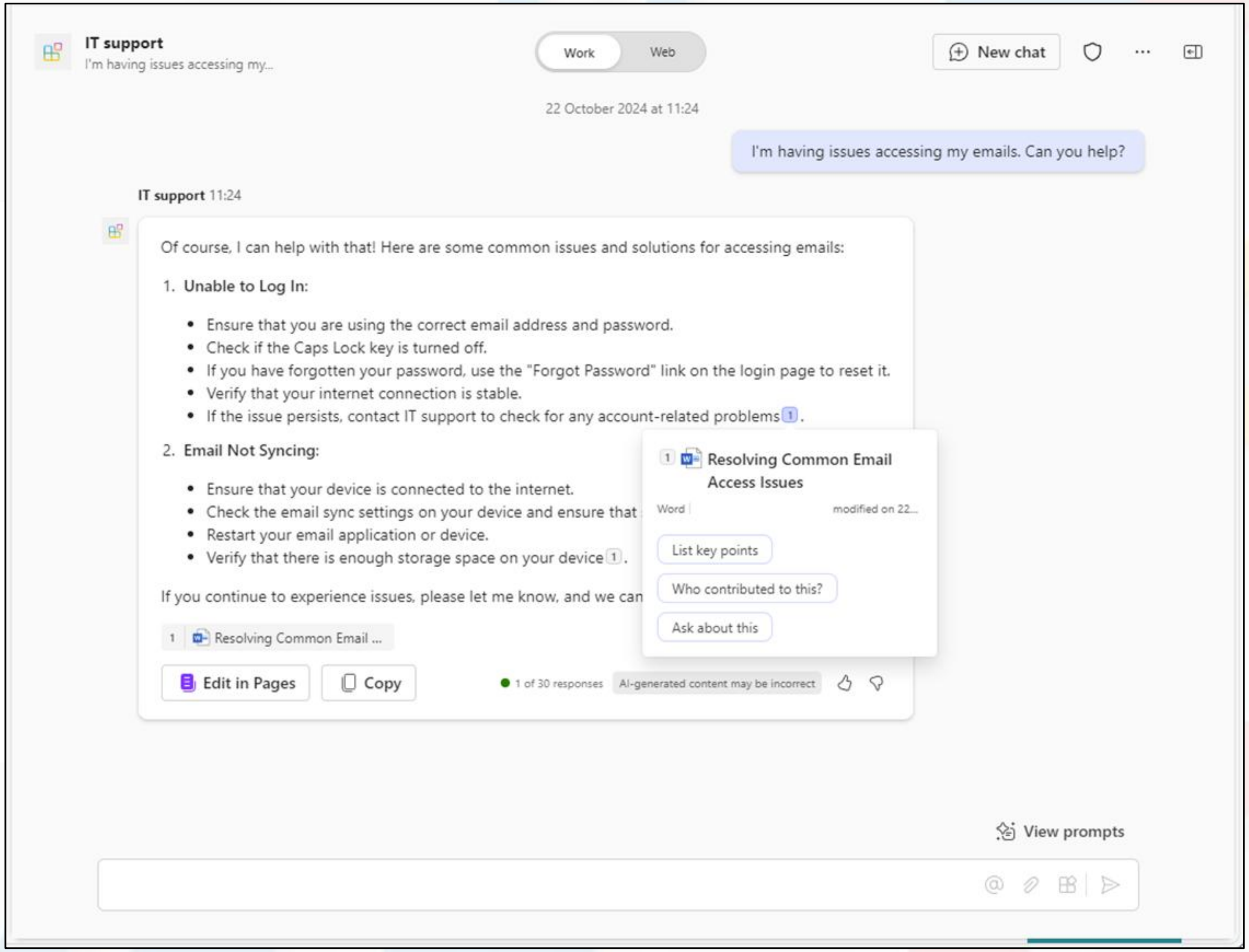
Generative answers =

Generative AI +

Knowledge grounding (RAG)

The AI does **NOT answer purely** from its own internal model knowledge unless you explicitly allow it.

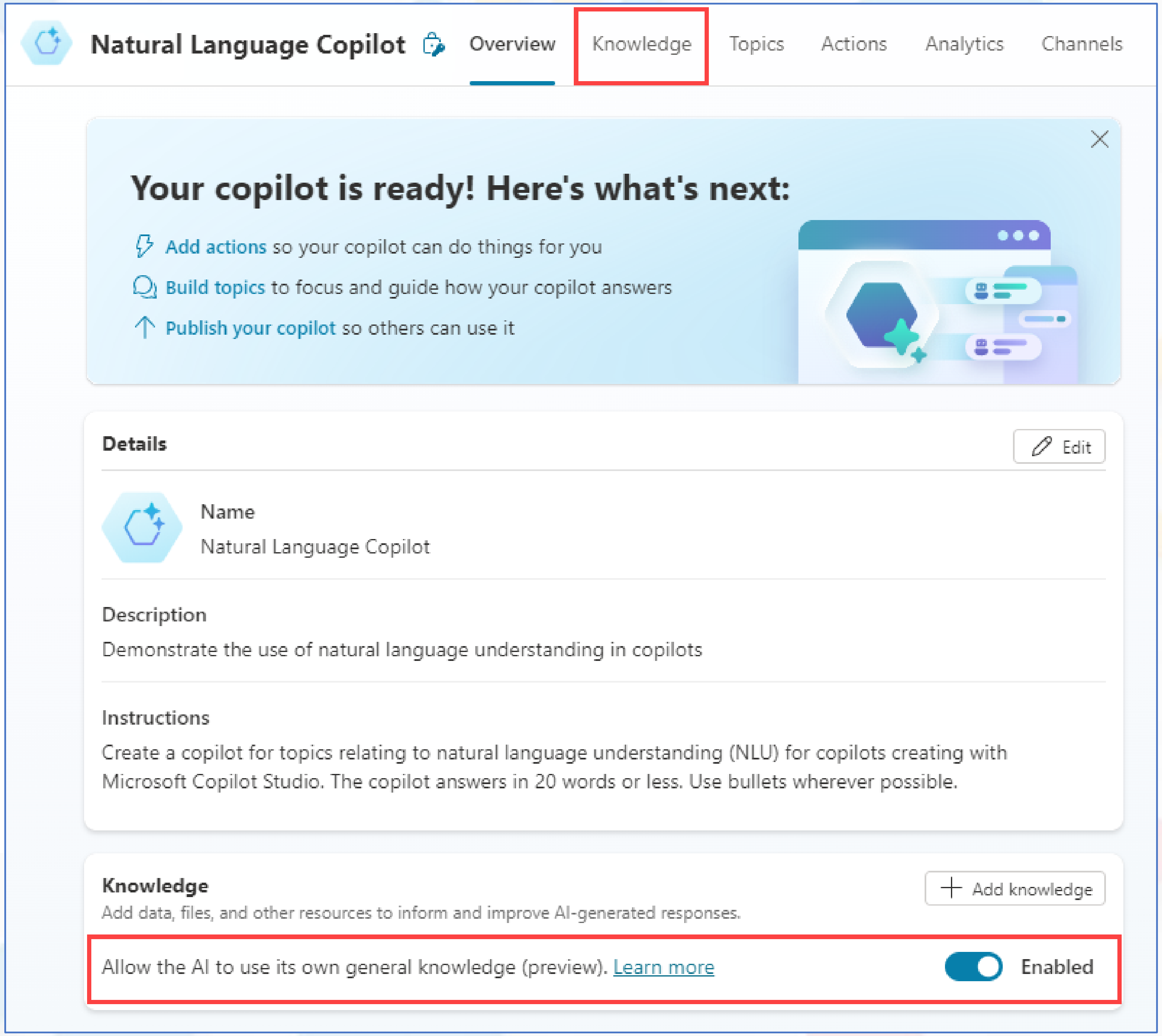
Add custom knowledge to your agent for more relevant information and insights.



AI General Knowledge

Enable AI to use its own general knowledge

- **Accessibility**: The agent can instantly access a vast repository of information and expertise across a wide range of subjects
- **Versatility**: It's capable of addressing diverse topics and tasks, making it a versatile resource for a variety of needs



Knowledge Sources For Generative Answers

The supported knowledge sources are:

Public Website

Searches the query input on Bing, only returns results from provided websites

SharePoint

Connects to a SharePoint URL, uses GraphSearch to return results

Dataverse

Connects to the connected Dataverse environment and uses retrieval-augmented generative technique in Dataverse to return results

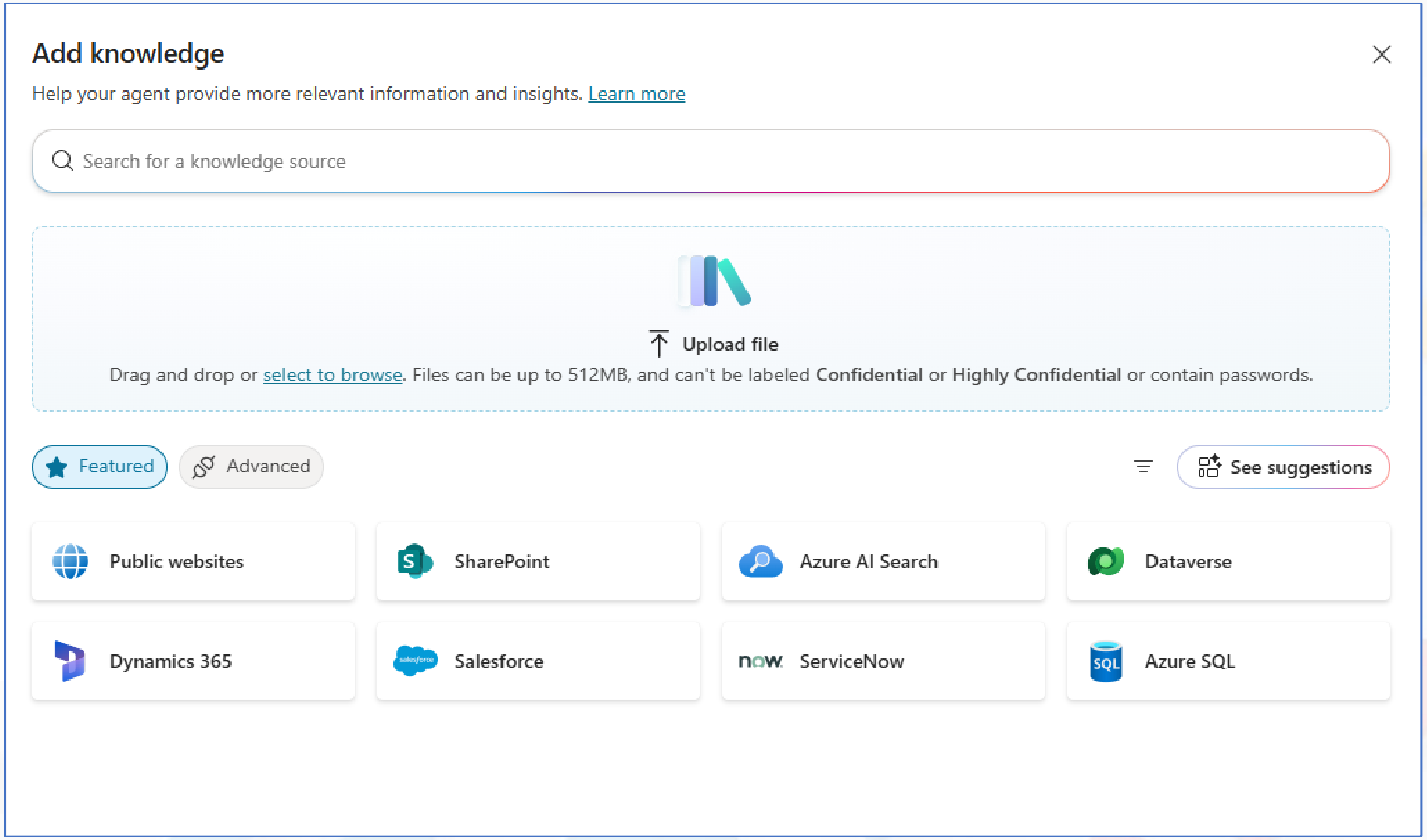
File Upload

Searches uploaded documents, returns results from the document contents

Knowledge Sources

Copilot Studio

- Content your agent can reference directly
- Supports generative answers grounded in organizational content
- Includes files, websites, SharePoint, Dataverse, and internal data



Using Websites As Knowledge Sources

URLs redirecting to another top-level site are not supported

For example, if www.fabrikam.com redirects to www.contoso.fabrikam.com, your agent will not generate responses

Websites that requires authentication or not indexed by Bing cannot be used.

For example, an internal site that require authentication like https://fab.visualstudio.com/project/wiki cannot be used.

URL can have up to two levels of depth

A trailing forward slash, however, is allowed

For example:

www.fabrikam.com/retail/stores/

Not:

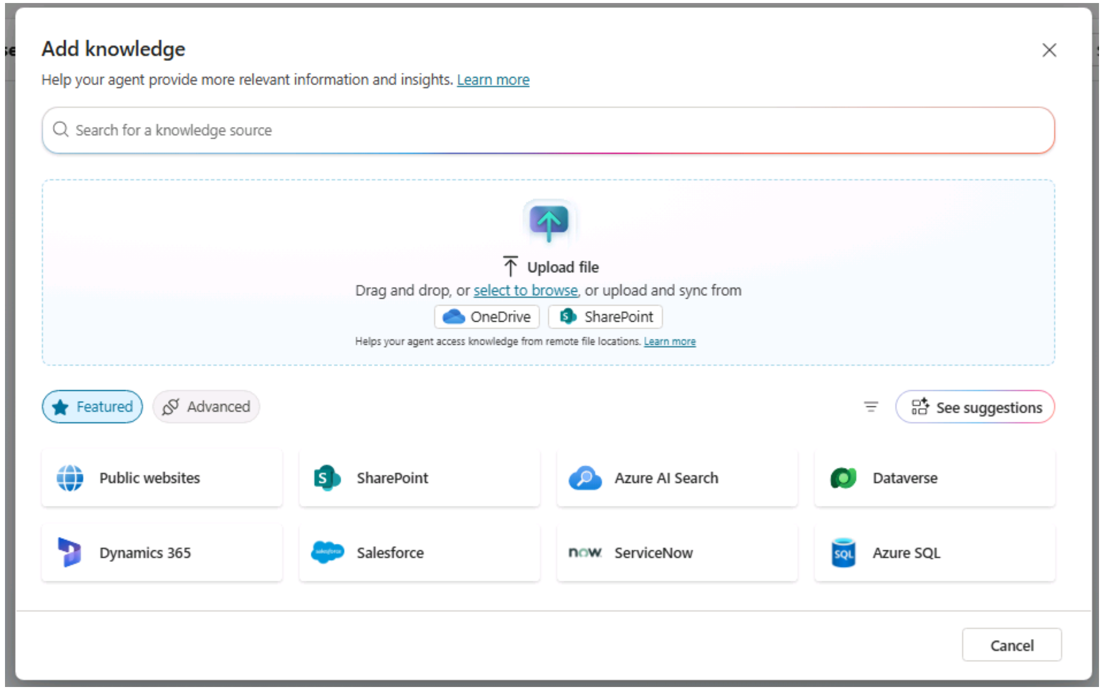
www.fabrikam.com/retail/stores/us

Using Documents As Knowledge Sources

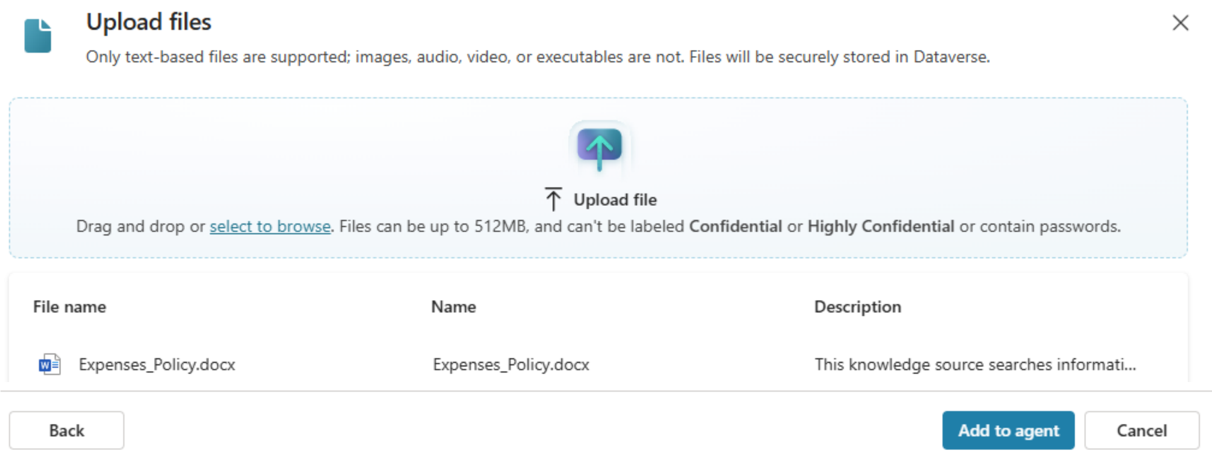
- Documents can be uploaded as a knowledge source for generative AI
- Documents are stored securely in **Dataverse**, and indexed for efficient search
- Document contents are visible to all agent users
- File size is limited to a max of 512 MB per upload
- Text-based files are recommended. Image, audio, video, and executable files are not supported

Upload the Document As Knowledge Sources

Select **+ Add knowledge** to see the multiple types of knowledge sources available.



In the **Upload file** section, use **select to browse** to upload the expense policy document you downloaded, then select **Add to agent**.



Add a Knowledge Source

Agent: **Expense Claim Assistant**

Add a document-based knowledge source for policy questions.

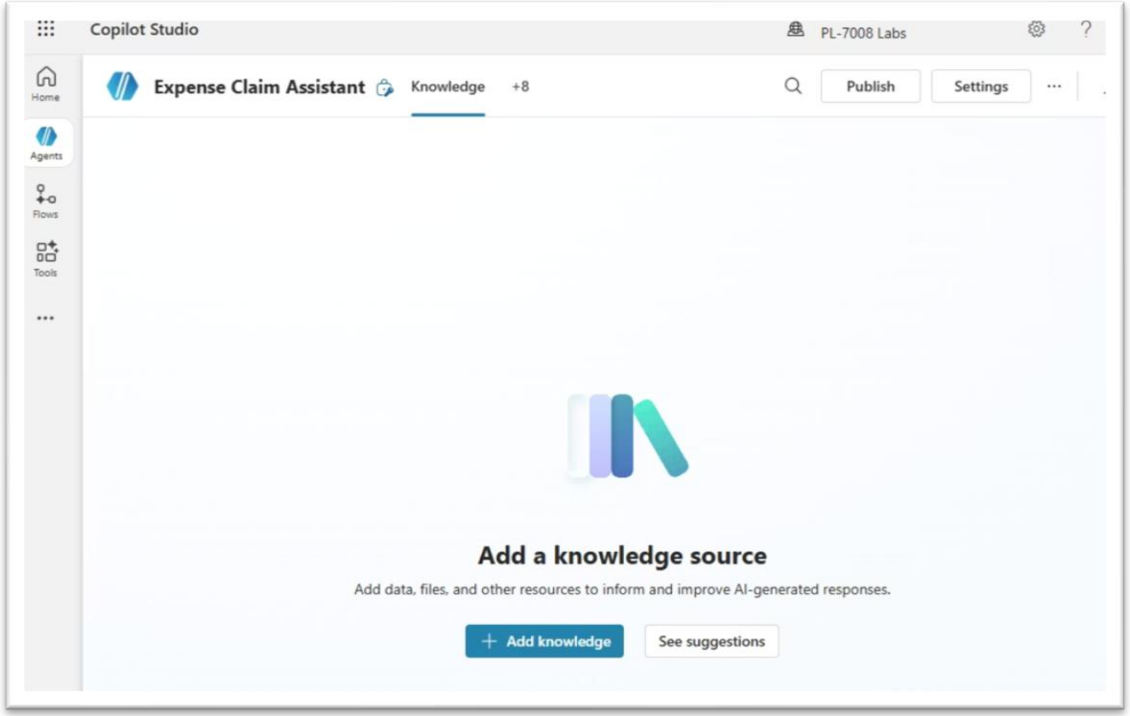
Download the Policy Document

Download **Expenses_Policy.docx** from the GitHub link (Contoso expenses policy).

https://github.com/MicrosoftLearning/mslearn-copilotstudio/raw/main/expenses/Expenses_Policy.docx

Open the Knowledge Tab

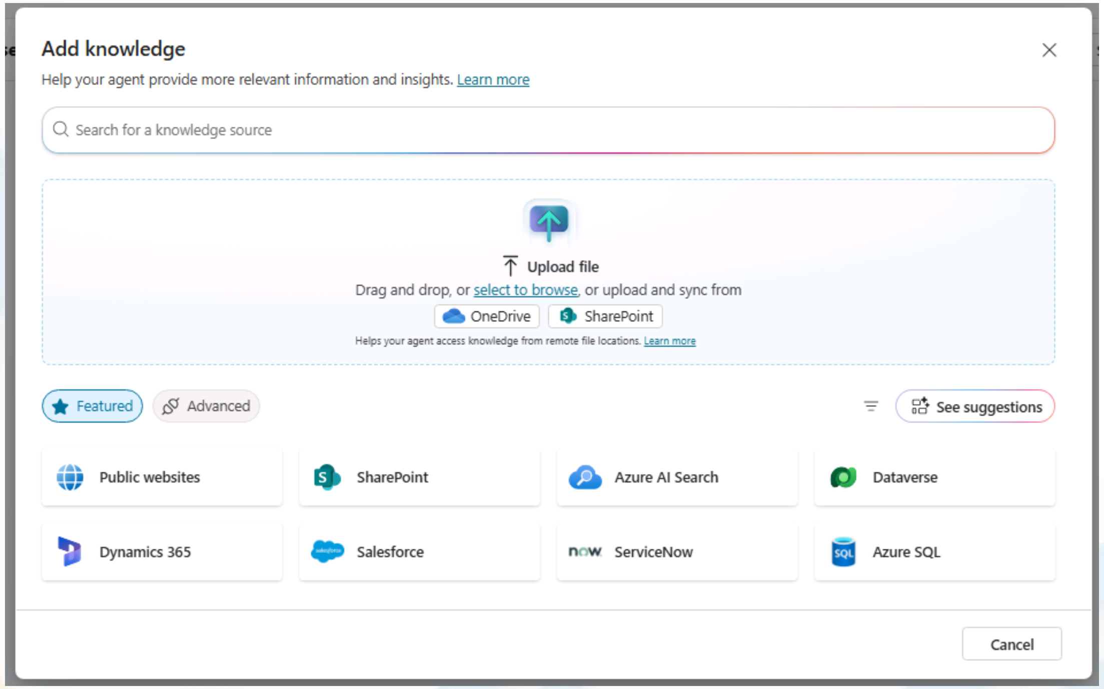
Close the Test pane and select the **Knowledge** tab (no sources should be listed yet).



Upload the Expenses Policy Document

Agent: **Expense Claim Assistant**

Select **+ Add knowledge** to see the multiple types of knowledge sources available.



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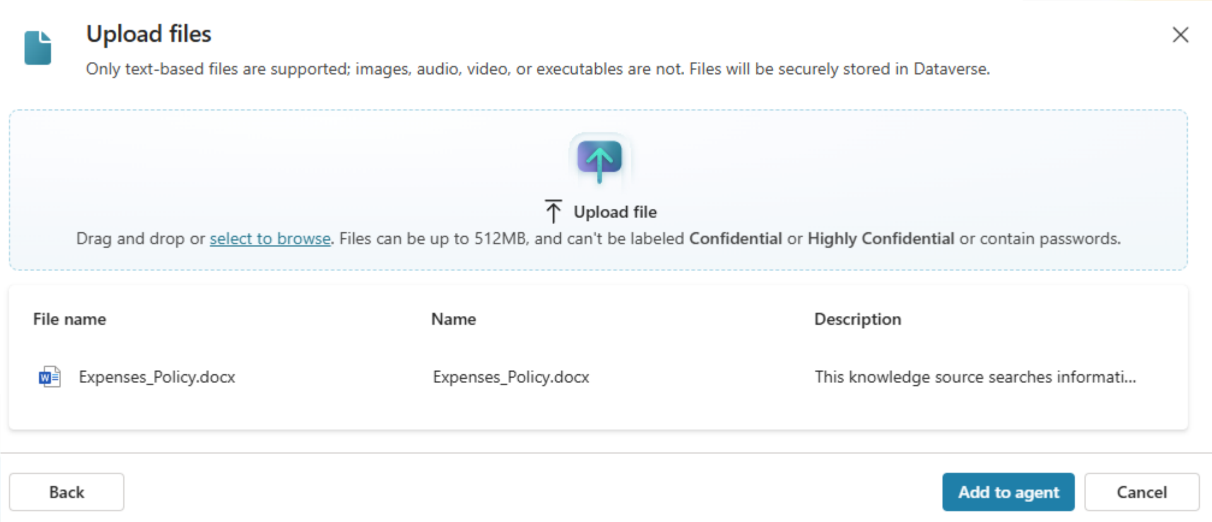
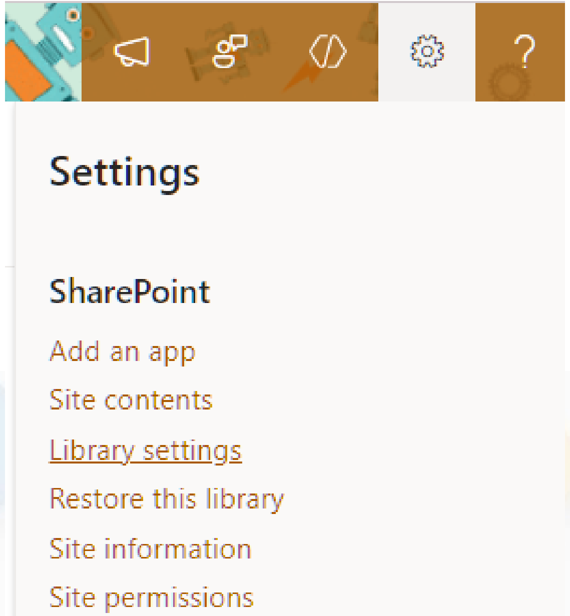


Add SharePoint As Custom Knowledge

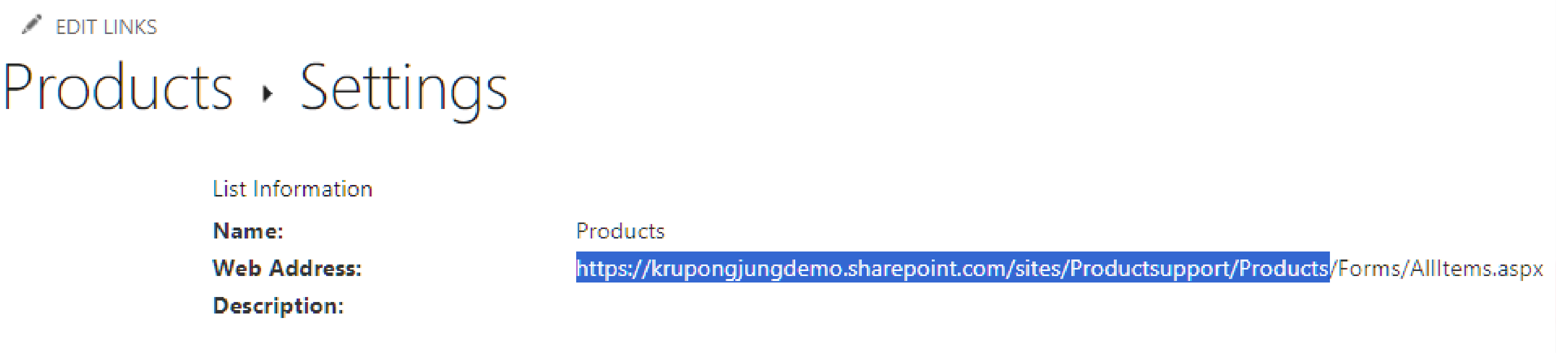
Copy the SharePoint URL

Next, copy the direct URL to the site for use when configuring your agent's knowledge.

1. From your SharePoint library page in SharePoint, select the **Settings** icon in the top right and choose **Library settings** then **More library settings**.



2. Locate the **Web address** property. Your **SharePoint site URL** is the portion of the Web address that is in the format `https://DOMAIN.sharepoint.com/sites/SITE_NAME/LIBRARY_NAME` where DOMAIN is your Microsoft 365 tenant domain.



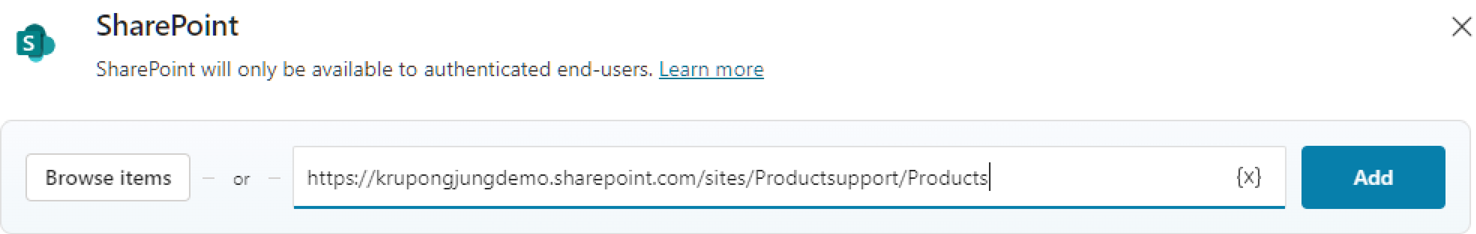
Do not include any portion of the URL that comes after name of your SharePoint library.

Add SharePoint As Custom Knowledge

Add SharePoint URL

1. In a web browser, navigate to **Microsoft Copilot Studio** at https://copilotstudio.microsoft.com.
2. Select **Agents**, and then, select your **Product Support** agent.
3. From the **Knowledge** section of the agent overview page, select **+Add Knowledge**.

4. On the **Add knowledge** page of the wizard that opens, select **SharePoint**.
5. In the text box, paste the URL of your **Products** SharePoint library then select **Add**. This should be in the format: `https://DOMAIN.sharepoint.com/sites/SITE_NAME/LIBRARY_NAME`

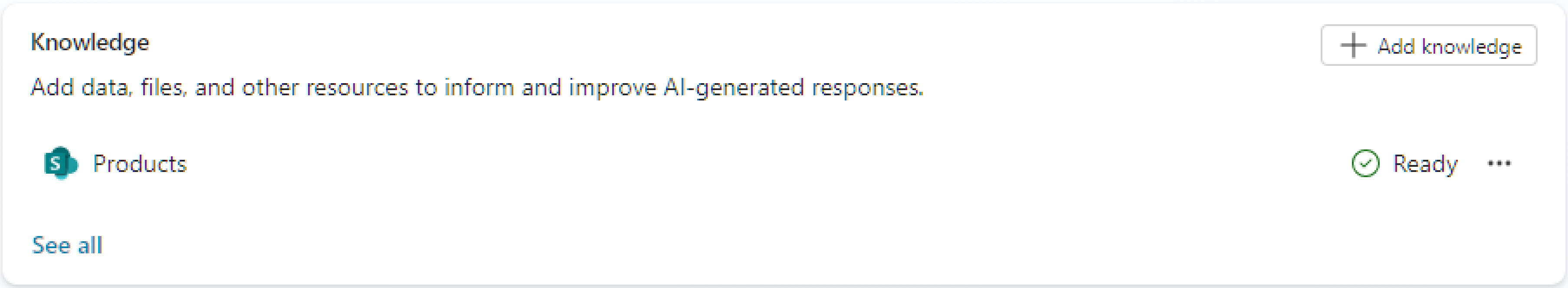


6. Select **Add to agent** then wait for the knowledge source to be added to the agent. This may take a minute.

Add SharePoint As Custom Knowledge

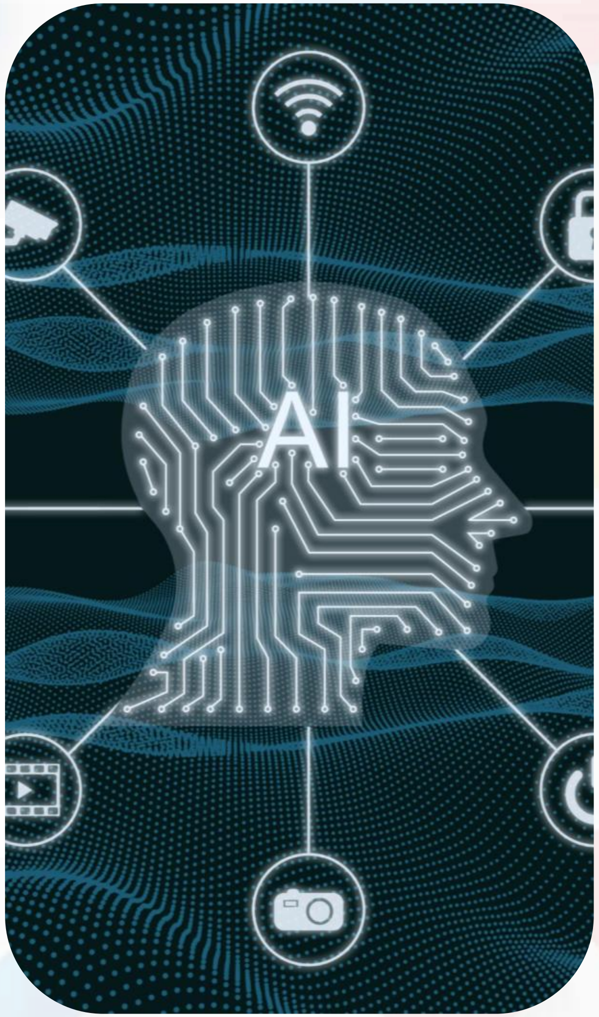
Add SharePoint URL

7. Notice that your SharePoint library is listed under the **Knowledge** section of the agent's overview information.



Designing Agents with Multiple Knowledge Sources

- Unified knowledge grounding
  - Microsoft Copilot Studio enables agents to generate responses grounded in **multiple knowledge sources within a single interaction** (one user message and one agent response).
  - Generative answers retrieve and synthesize information across internal and external sources
- Flexible knowledge configuration
  - Knowledge sources can be configured at the **agent level** or within a **topic level**
  - All configured sources are applied together to ground responses at runtime
- Simplified agent design
  - Reduces the need for extensive manual topic authoring
  - Enables agents to handle a broader range of user questions using existing knowledge



Module 5: Tools and Actions (Overview)

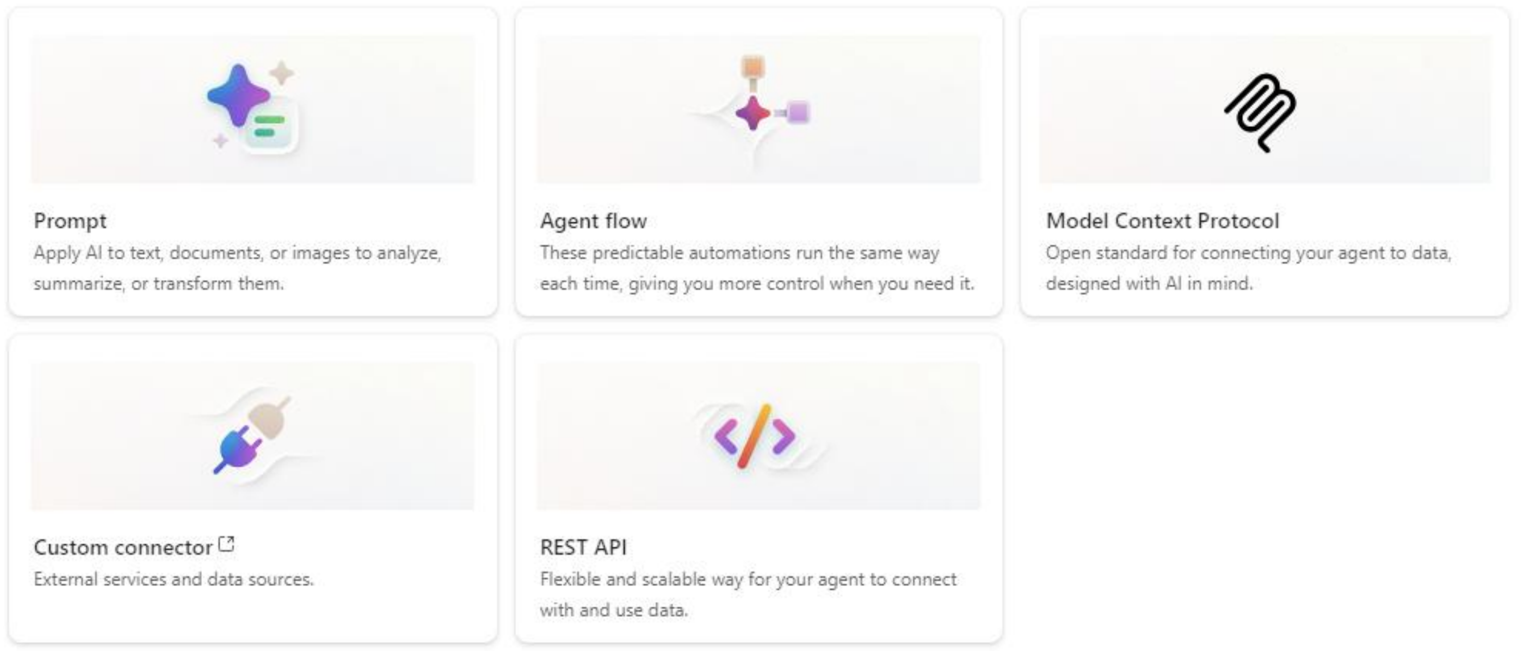
This module provides an overview of agent tools and actions, demonstrating how agents call tools, integrate with Power Automate, use variables and conditions, and orchestrate simple workflows, with a demo showing an agent invoking a Power Automate flow as a tool.

Tools in Copilot Studio

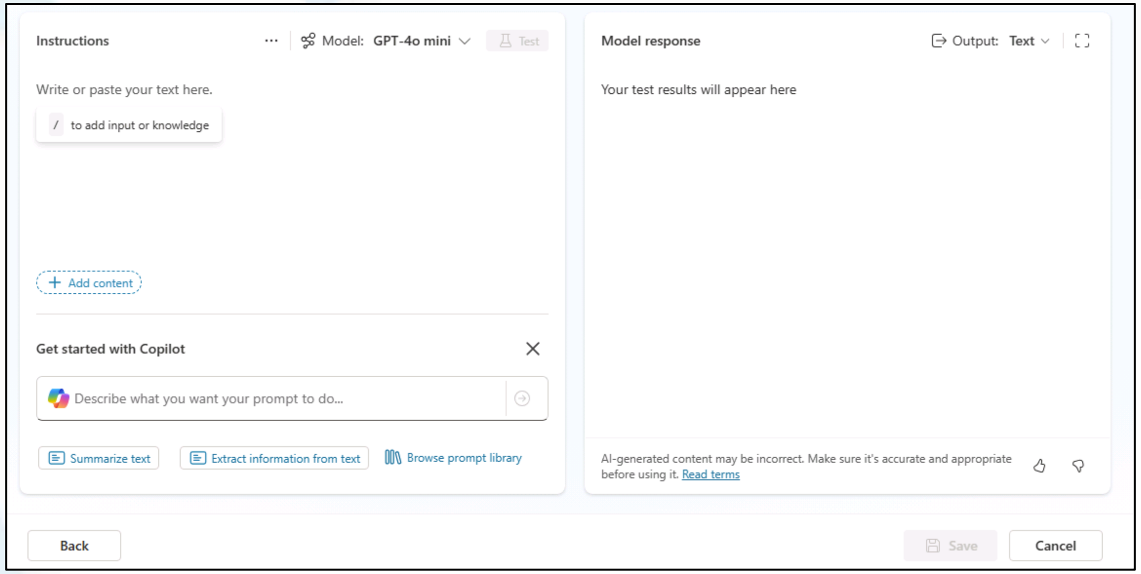
- **Reusable capabilities** within a Copilot Studio environment
- Core tool remains the same – wrapper around tool allows for customization of inputs, outputs, and configurations per agent
- Provide **specific features or tasks** to an agent
- Find, manage, and create tools on the **Tools** page

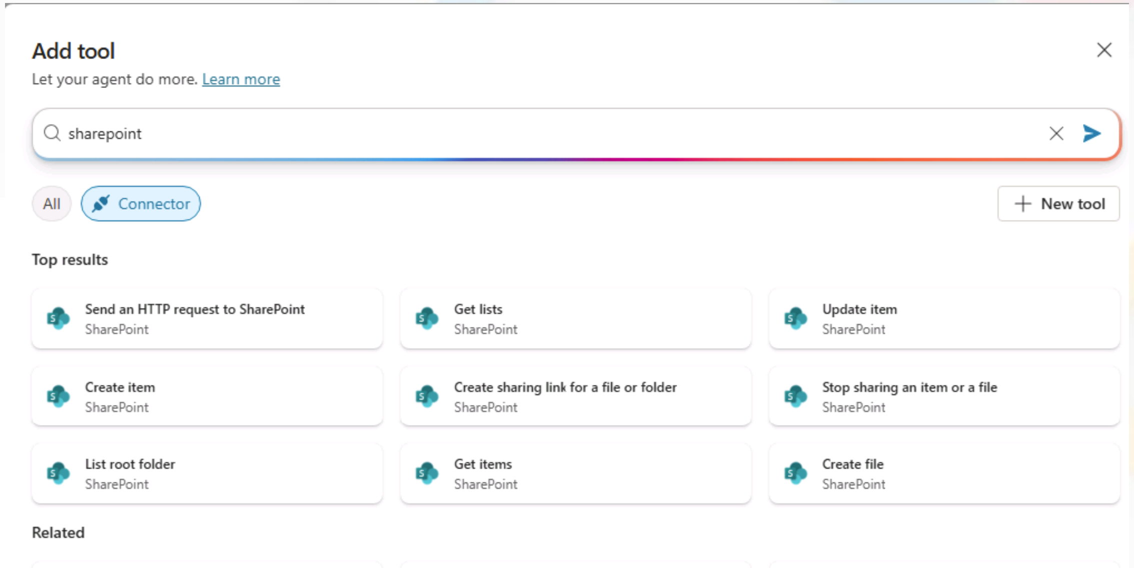
Types Of Agent Tools

- **Prompt**: Run pre-defined prompts to customize the generation of text content in relation to user input and related knowledge (Dataverse).
- **Connector**: Retrieve and update data from external sources through APIs to create more dynamic and useful agents
- **REST API**: Upload Swagger files to connect agents with external data sources or services not covered by prebuilt connectors (Preview)
- **Model Context Protocol (MCP)**: Connect agents with existing knowledge servers and data sources to improve the use of structured and unstructured data
- **Agent Flow**: Enable agents to automate deterministic workflows. (Not available to declarative agents)



Types Of Agent Tools





Prompt tools

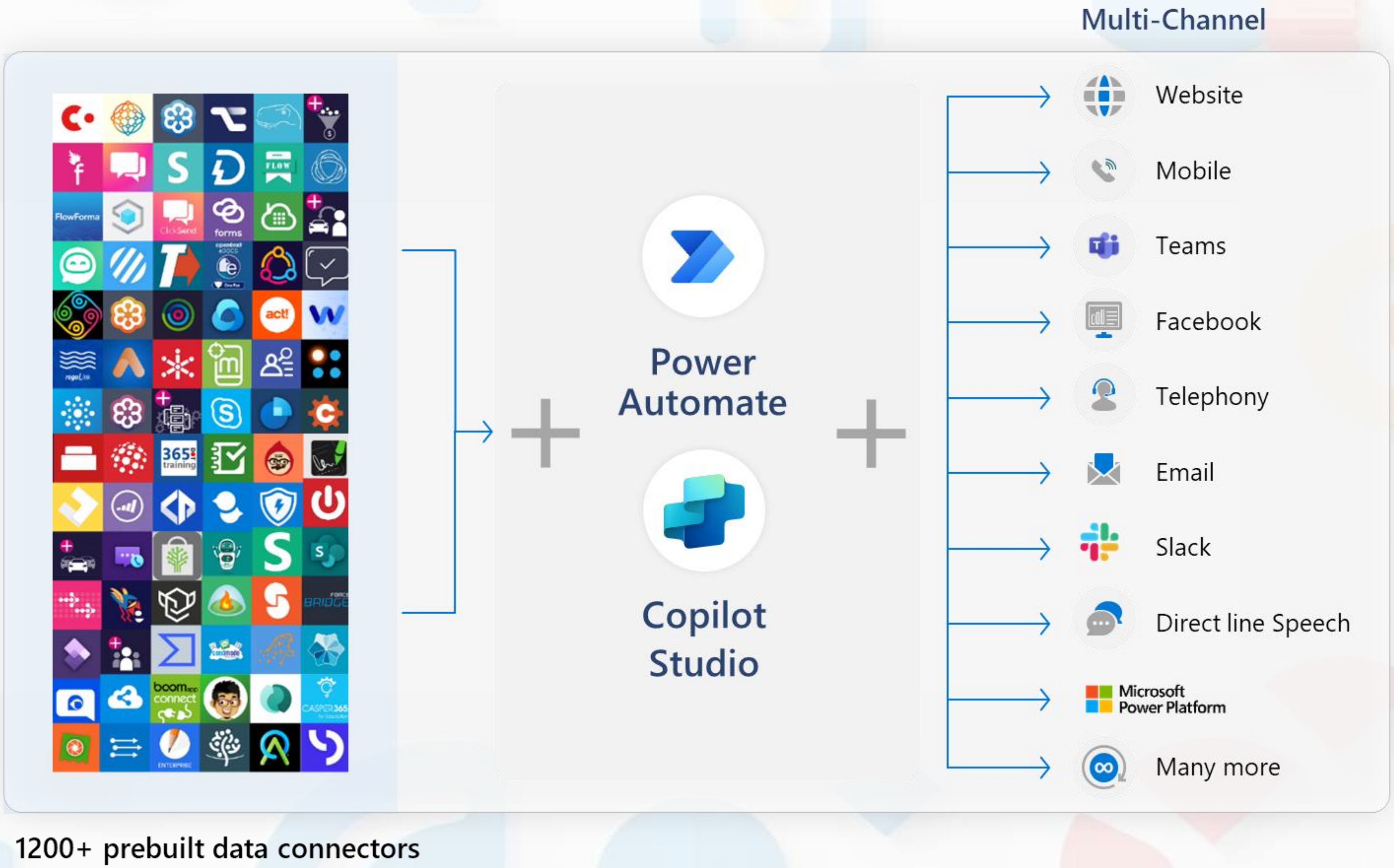
- Run pre-defined prompts
- Customize the generation of text content in relation to user input
- Leverage knowledge (Dataverse) and user input in prompts

Connector tools

- Retrieve and update data from external sources through APIs
- Create more dynamic and useful agents
- Use existing Microsoft certified connectors + custom connectors

Power Platform Connectors

- Power Platform, including Power Automate and Copilot Studio, offers great integration capabilities, with more 1,200 native connectors or ways to create your own custom connectors to your APIs.
- For a good end-user experience, cloud flows, HTTP requests and connectors triggered from Copilot Studio must execute quickly so that the user doesn't have to wait too long for the agent to answer.



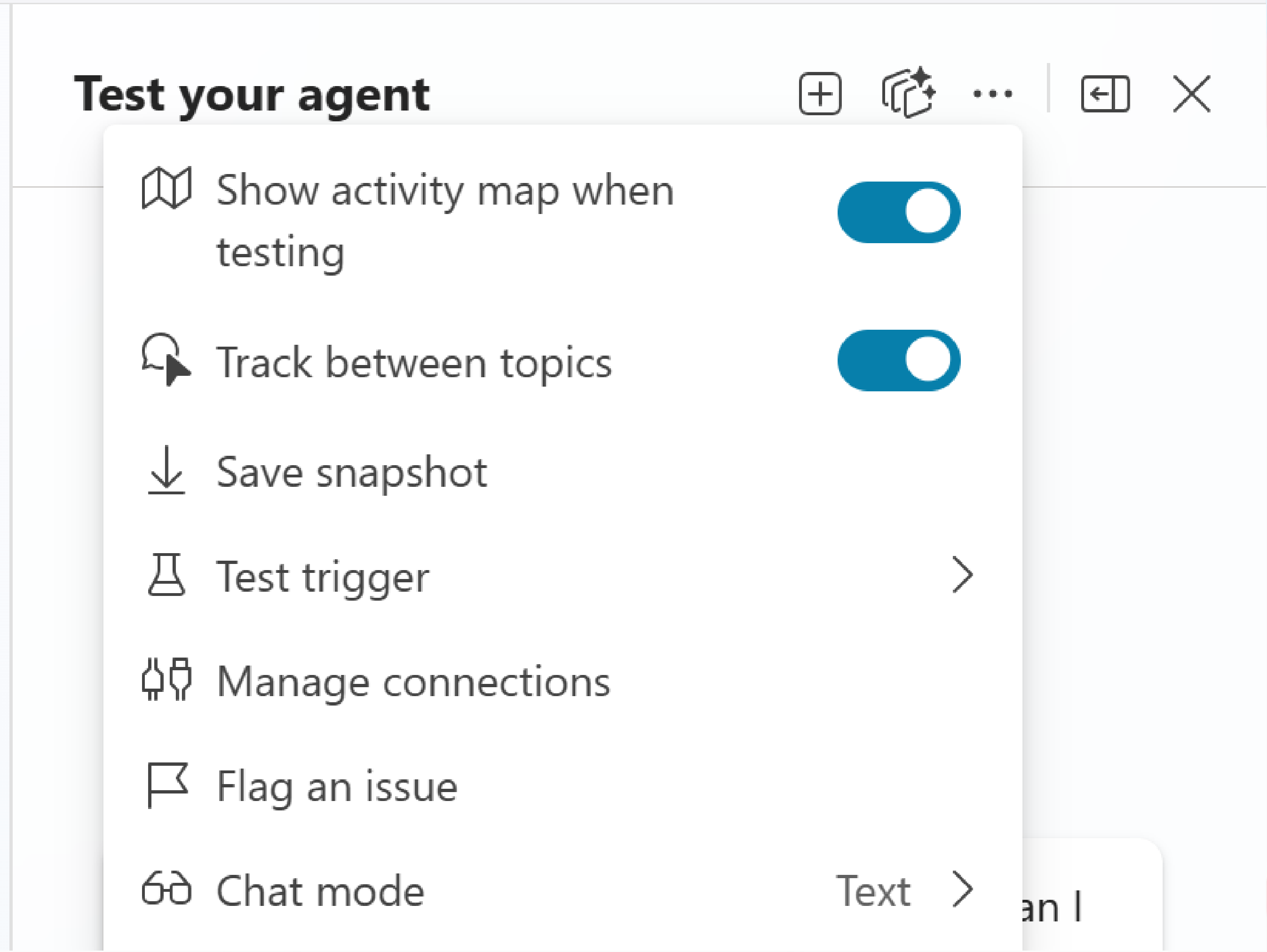
Module 6: Testing, Publishing and Monitoring

This module covers how to test, debug, refine, publish, and monitor agent conversations, introducing key security and governance concepts along with an overview of analytics to ensure reliable, secure, and well-governed agent behavior.

Test agent

Test panel options

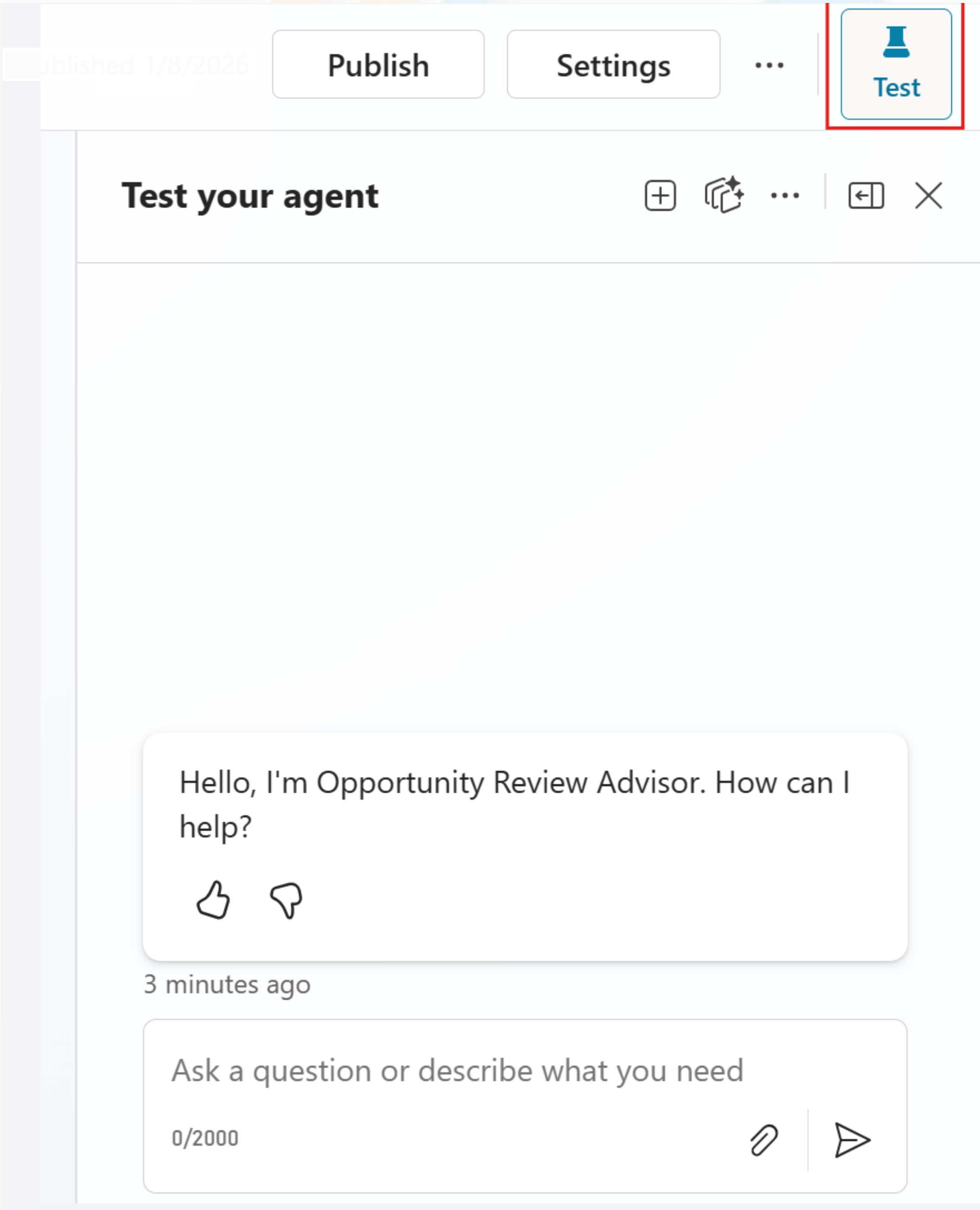
- Activity map
- Track between topics
- Start a new test session



Test panel

Show and hide by selecting Test button

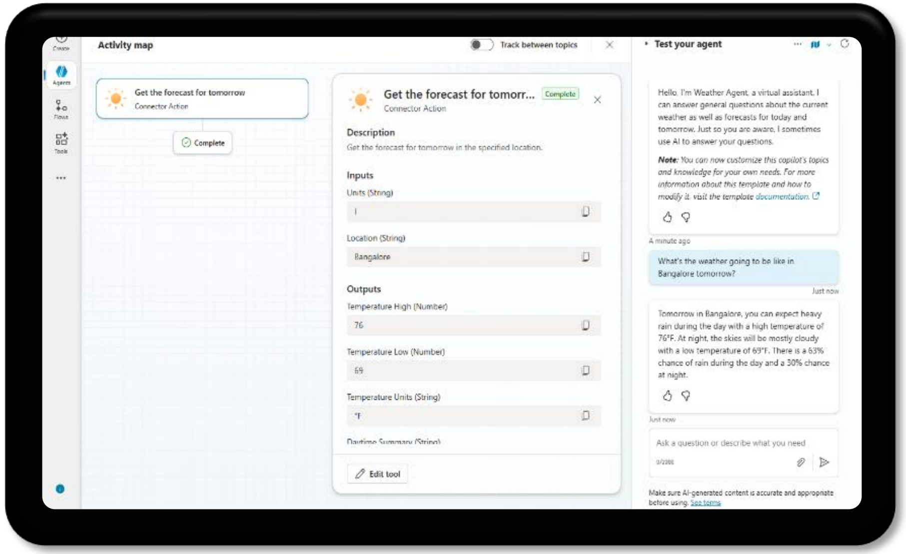
- Ask questions or describe what you need
- View the results



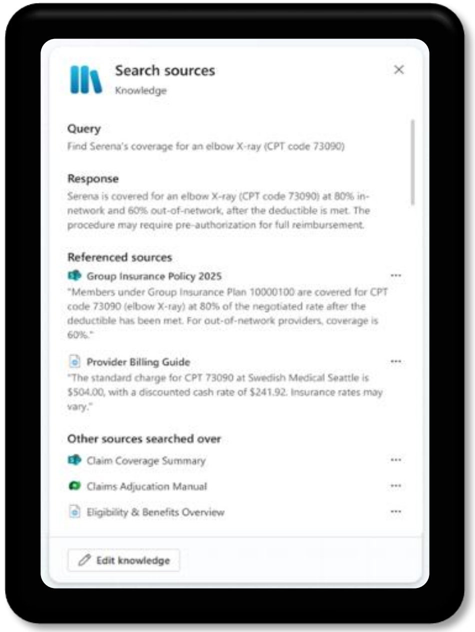
Easily Keep Track Of Your Agent's Activity

Review real-time and historical records of your agent's sessions to ensure your agent is meeting your goals

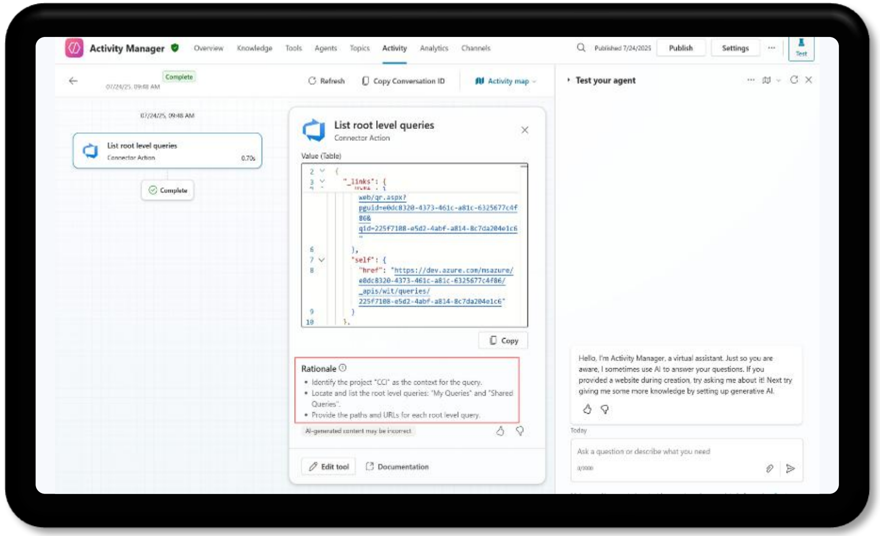
Activity map



Details



Rationale



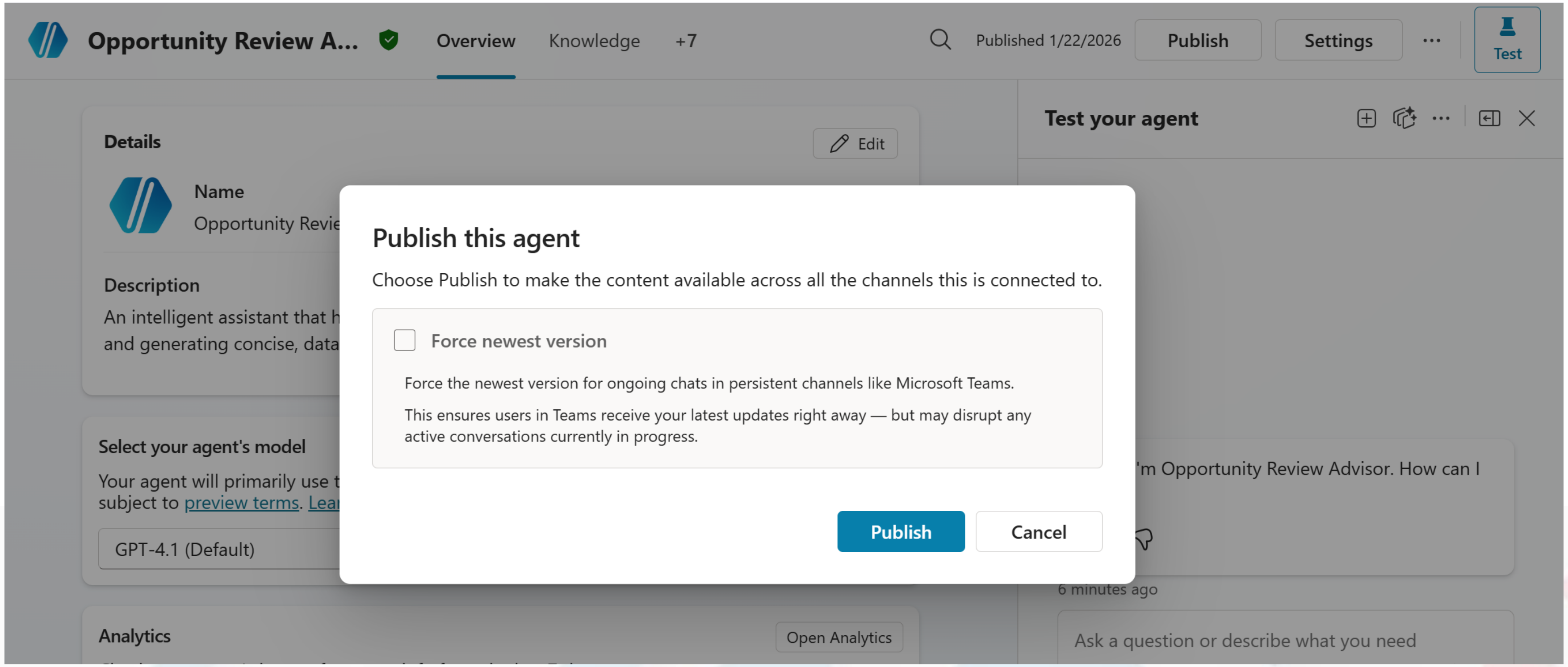
Quickly troubleshoot by reviewing auto-generated explanations for why your agent decided to call a particular tool or fill in parameters.

Publish Agents To Microsoft 365 Copilot And Microsoft Teams

Availability option	Description
Share Link	• Share a deep link into Microsoft Teams that will invoke the published agent. • Opening the link in the tenant opens a chat experience with the agent.
Show to my teammates and shared users	• Grant access to others to participate in authoring the agent, or to security groups to grant them access to use the agent in Microsoft 365 Chat or Teams.
Show to everyone in my org	• Submit to the tenant admin to add the agent to the organizational catalog. • Tenant users can install as desired.
Download as a .zip	• Download as a zip folder. • You can manually upload to Teams/Microsoft 365 Copilot to submit to the admin for review or upload directly to the organizational catalog.

Publish

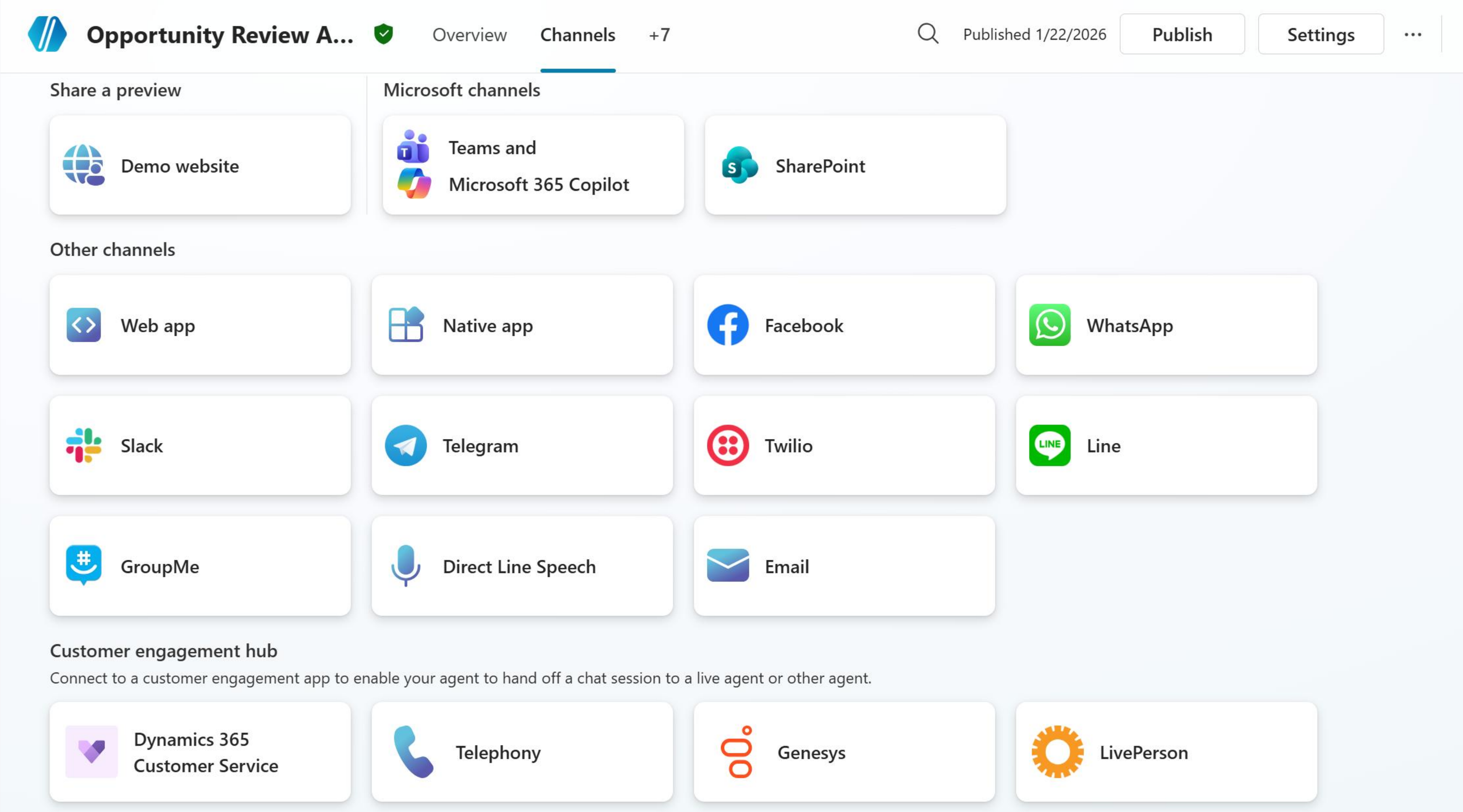
- Share with others
- Demo website
- Make the agent available on channels



Teams channel

Deploy to Microsoft Teams

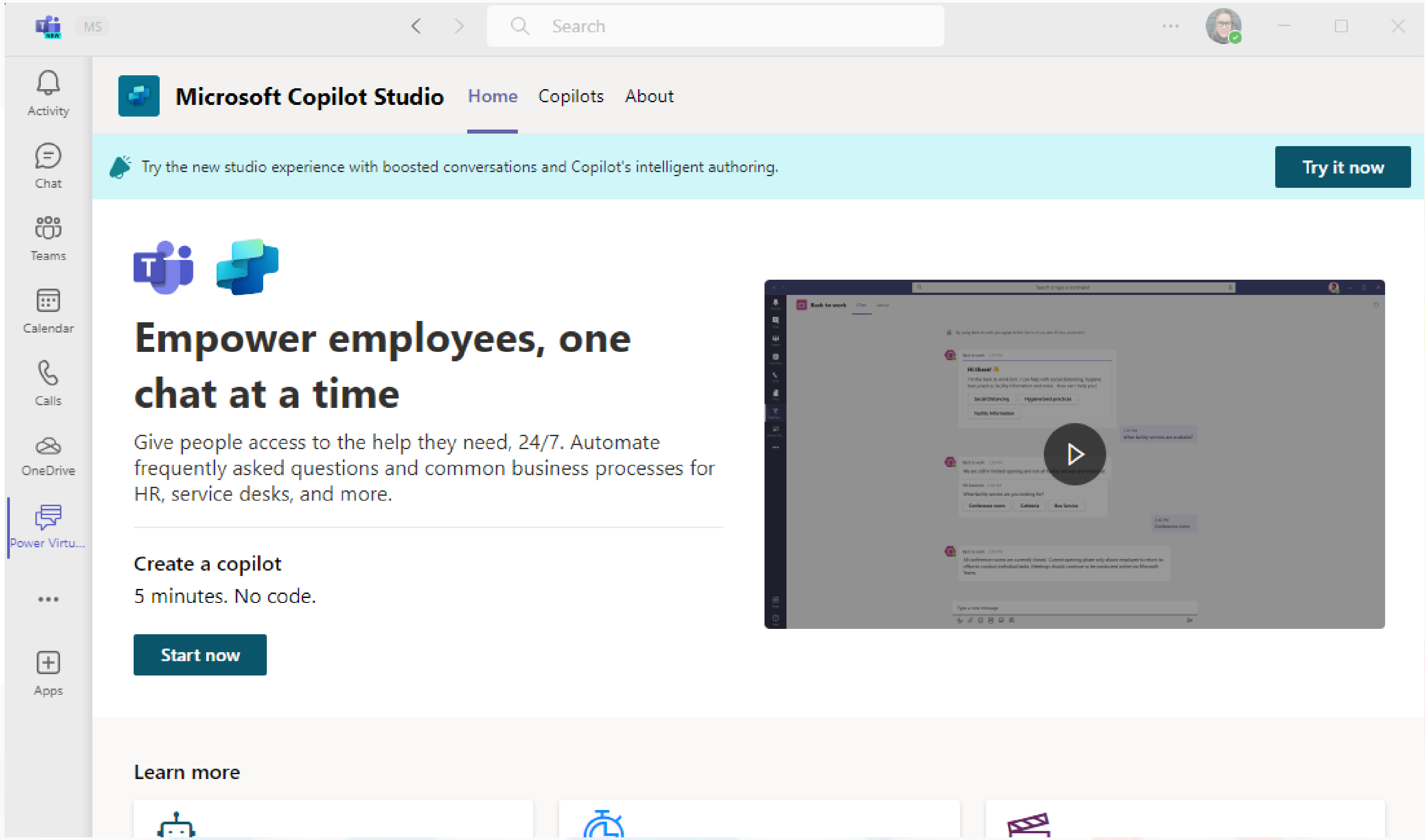
- Options to control the appearance of the agent
- Controls how a user can install the agent into a Microsoft Teams team



Install the Copilot Studio app in Microsoft Teams

Teams app store

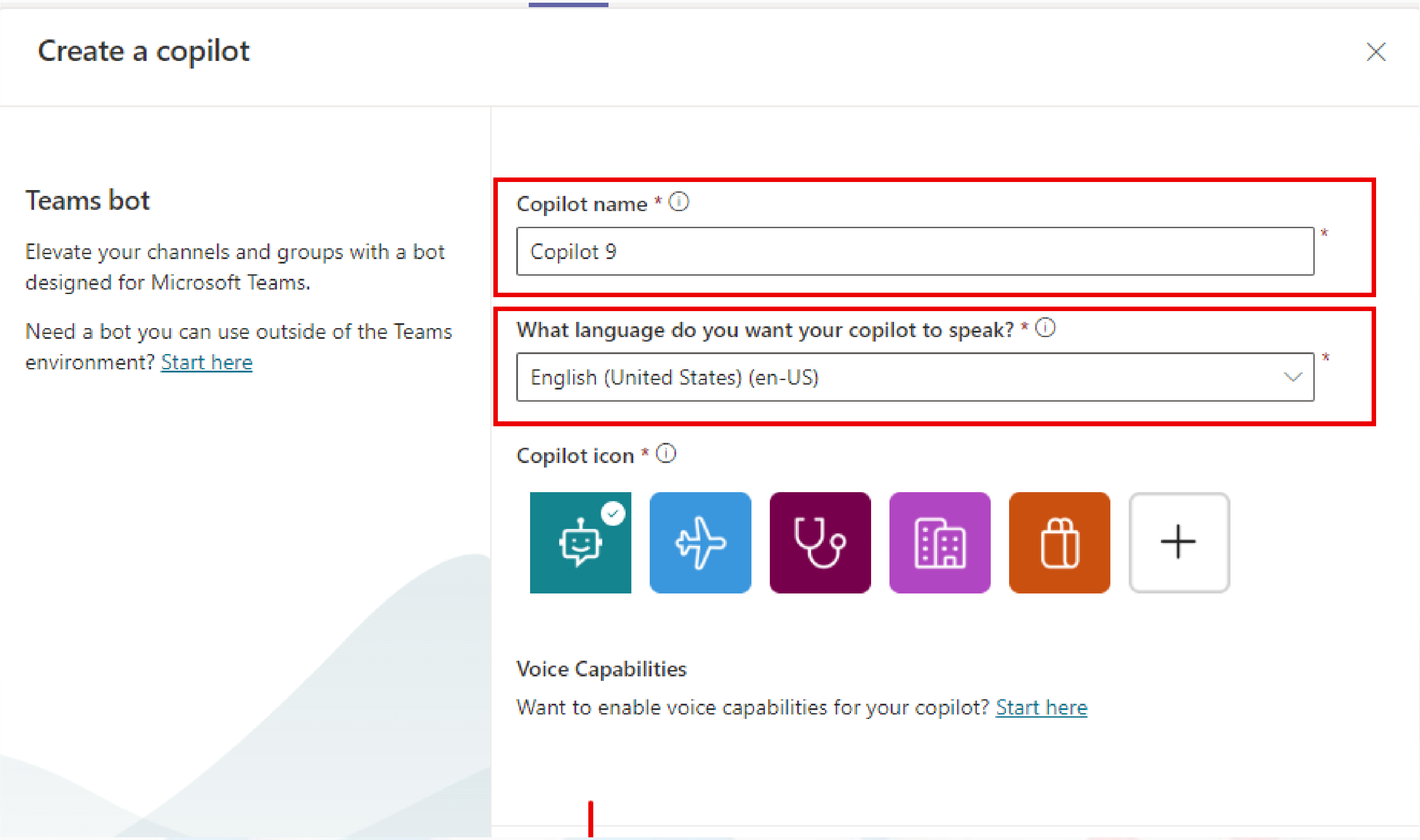
- Install app in Teams
- Pin to left-hand rail



Create an agent in Microsoft Teams

Copilot Studio app in Microsoft Teams

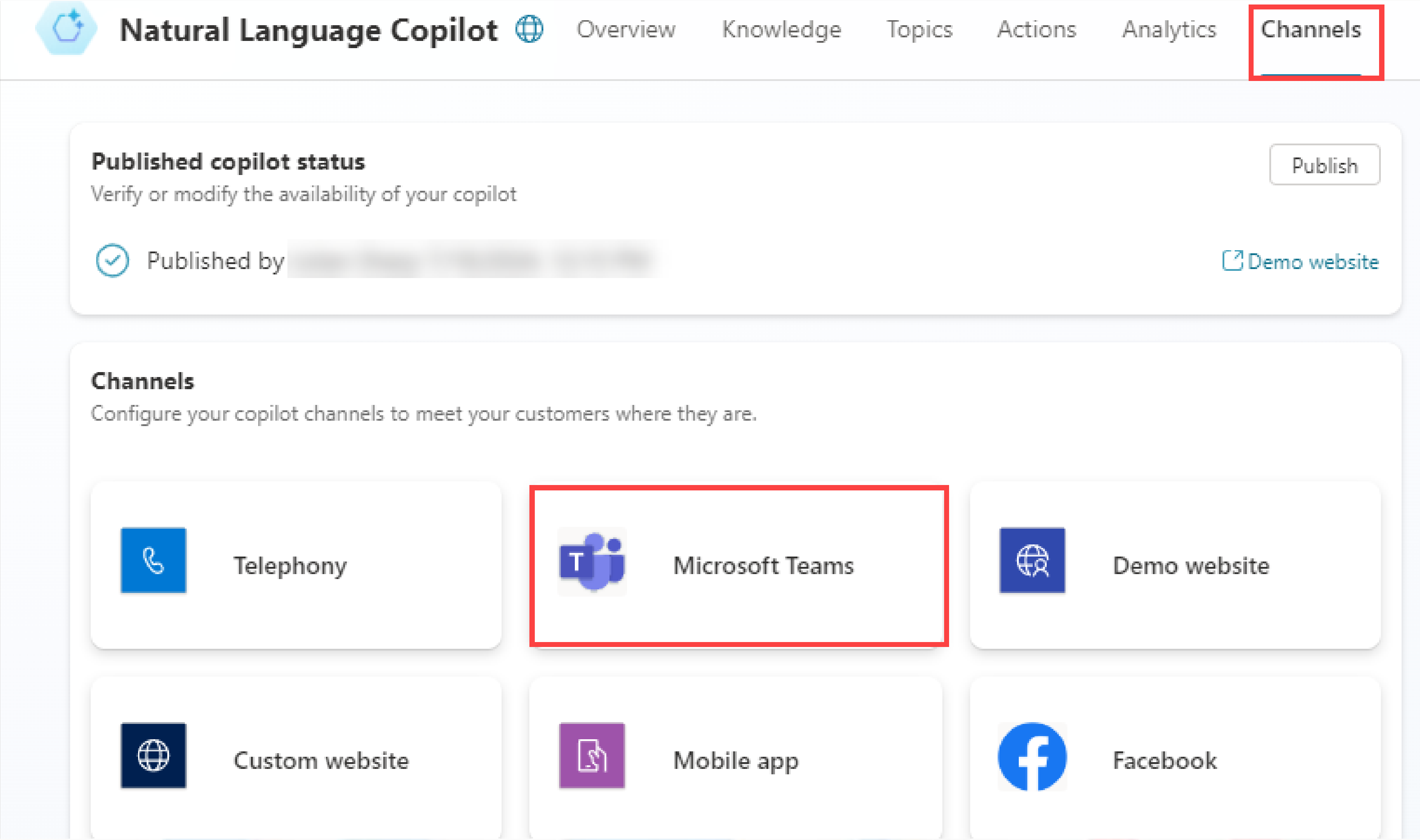
- Select the Team to own and manage the agent
- Deployment of the agent is limited to the users in the Team
- Team members only have read access to the agent



Channels

Deploy to Microsoft Teams

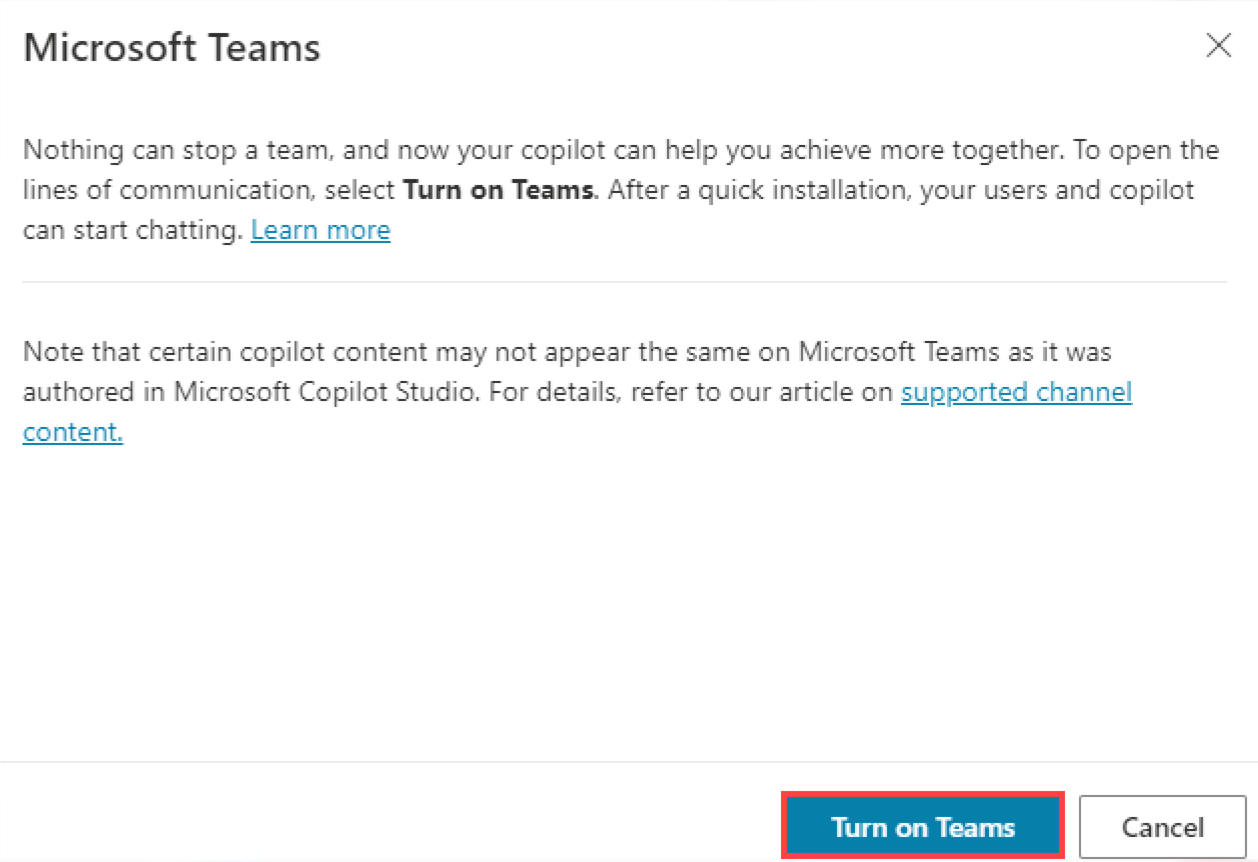
- Options to control the appearance of the agent
- Controls how a user can install the agent into a Microsoft Teams team



Teams Channel

Turn on Teams

- Connects agent to Teams



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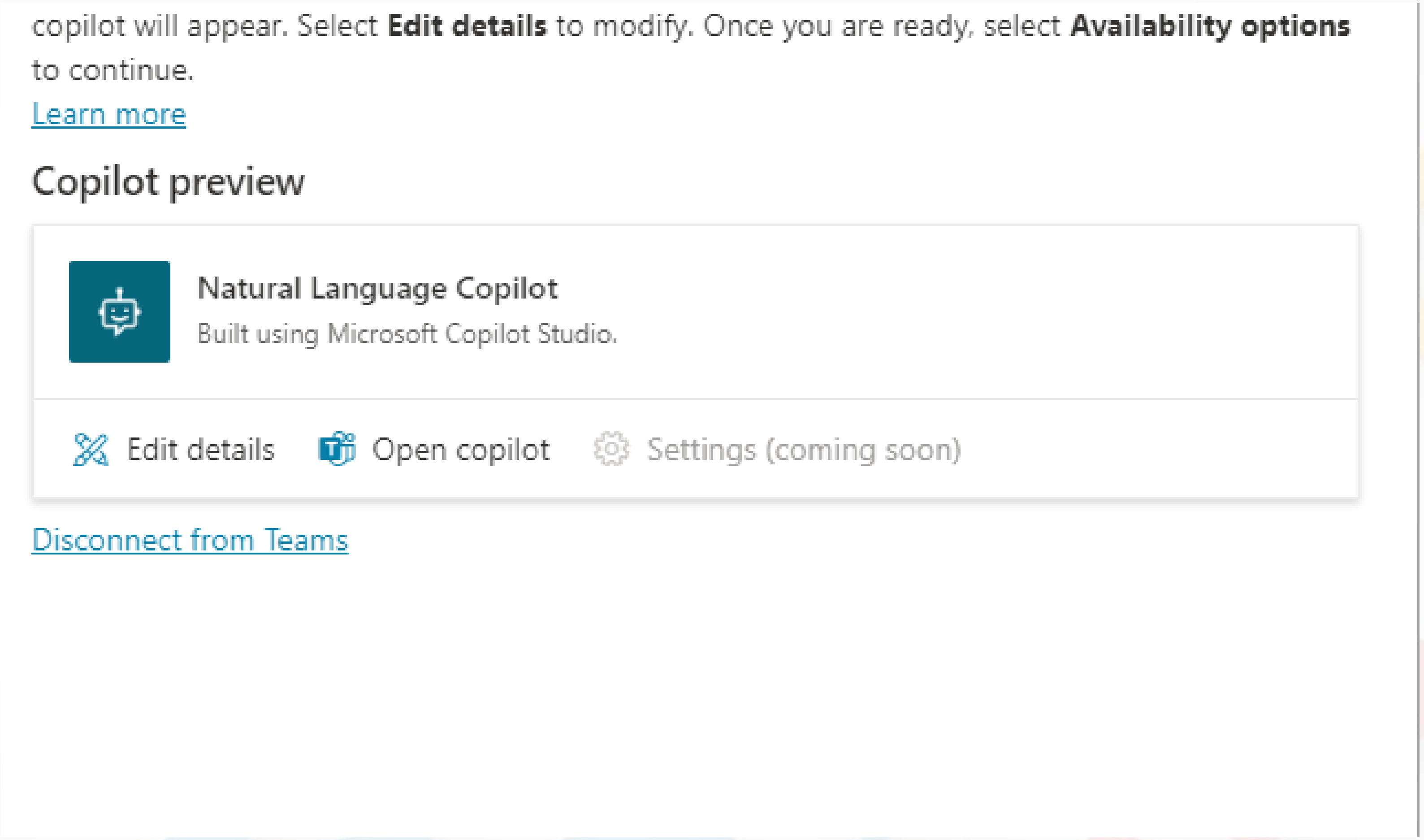
Agent Added To Teams

Customize the appearance of the agent

- Icon
- Color
- Description



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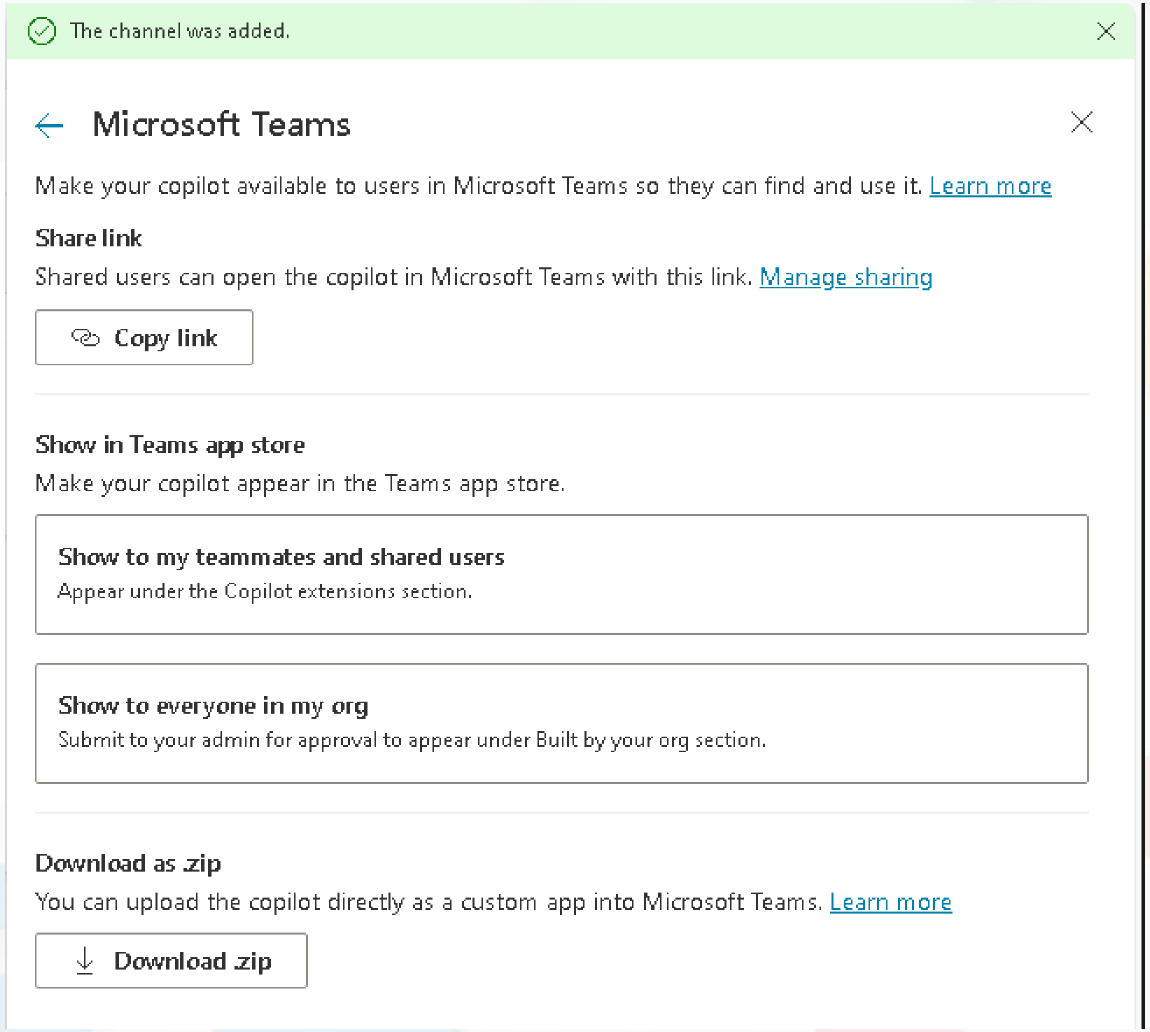
Options For Installing An Agent Into A Microsoft Teams Team

Availability options

- Share a link
- Show an agent in the Teams app store
- Download a zip file



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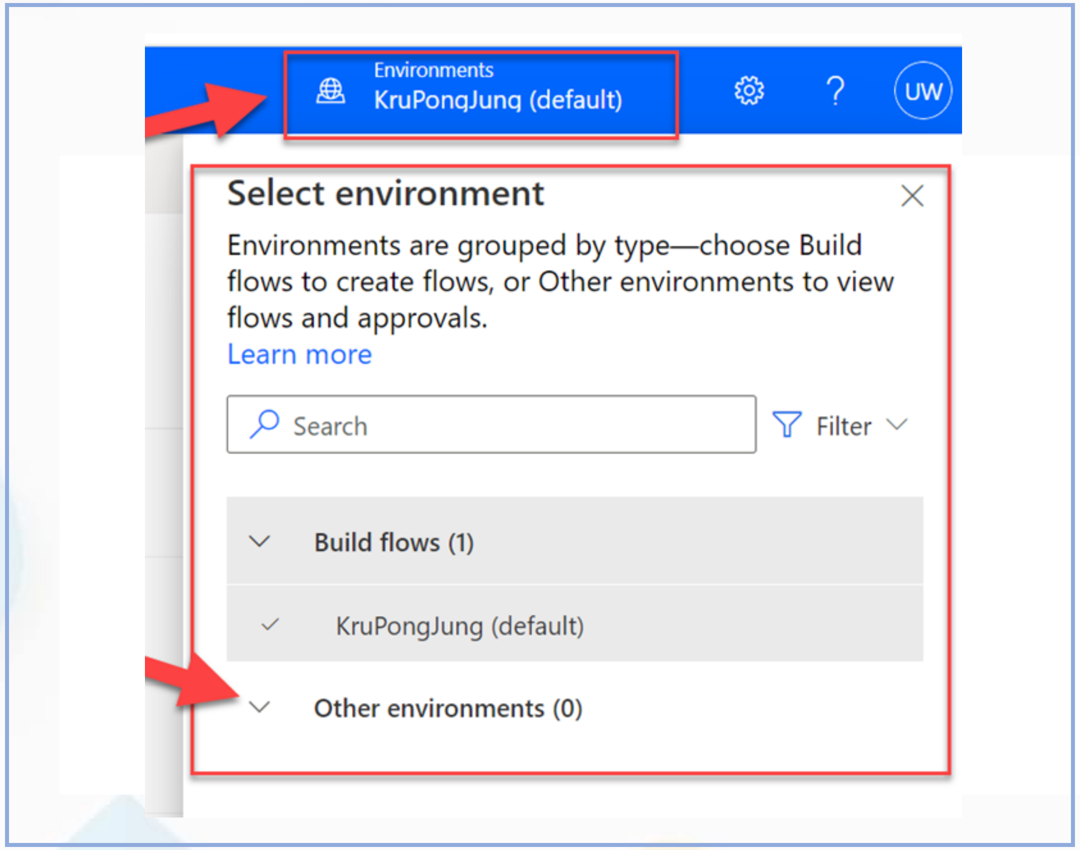
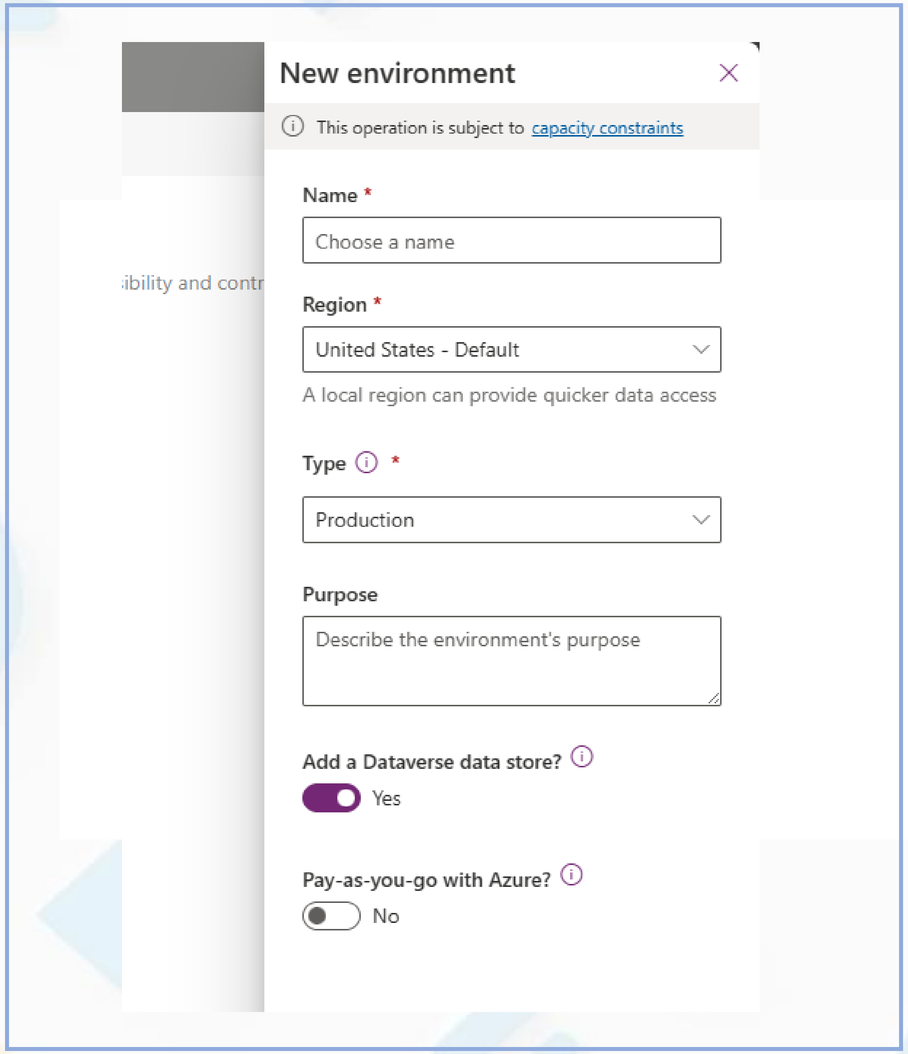
Environments In Microsoft Copilot Studio



- Environments help manage and share business data, including agents, apps, and flows, effectively.
- Environments can have different roles, security requirements, and target audiences.
- Organizations may decide to align their environment strategy by departments or by location.
- Global companies need multiple environments for data privacy.

Understanding Default Environments in Copilot Studio



- A **default environment** is created upon signing in for the first time.
- All subsequent agents are created in the default environment unless specified otherwise.
- Additional environments can be added for various needs and regions.
- Manage your environments in the **Power Platform admin center**.

Environment strategy

Having access to Copilot Studio ≠ **Being able to author agents on any environment**

- To author agents, users must be assigned a **security role** on the environment.
- The recommended role to author agents is **Environment Maker**.
- The **default environments** grants all users the Environment Maker role.
- While you can’t prevent users with access to Copilot Studio to create agents in the default environment, there are ways to **redirect users to their own personal developer environment**, and also to **block publishing of these agents** in the desired environments.

Microsoft

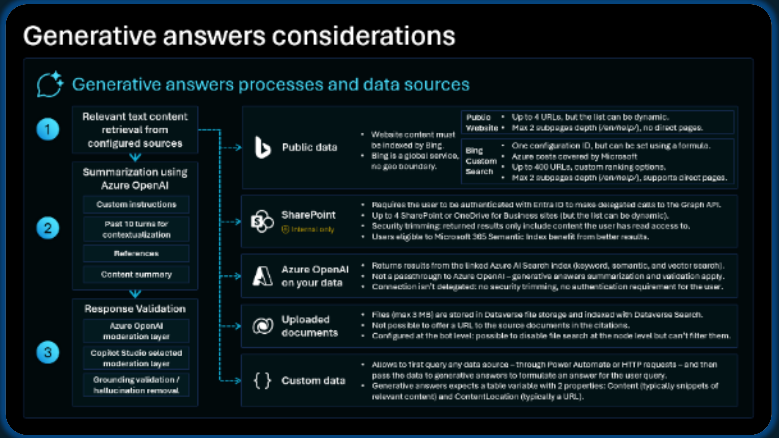
Copilot Studio Implementation Guide

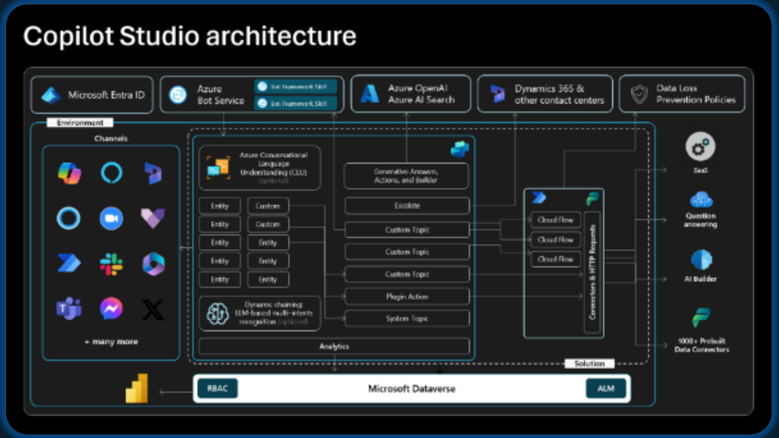
The Copilot Studio implementation guide provides a framework to do a 360-degree review of a agent or bot project.

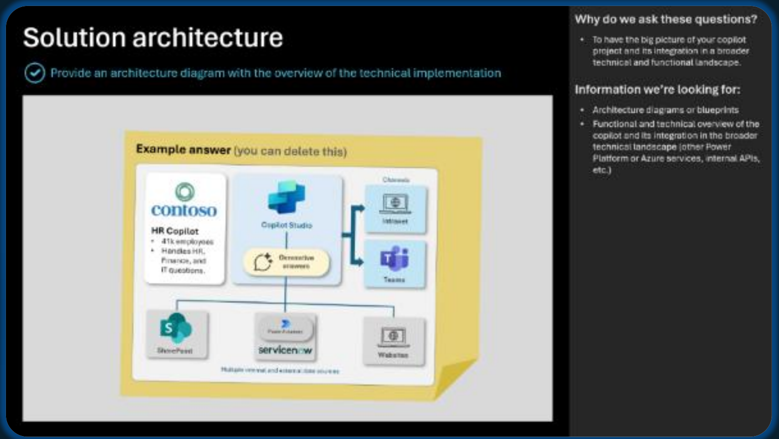
Through probing questions, it highlights potential risks and gaps, aims at aligning the project with the product roadmap, and shares guidance, best practices and reference architecture examples.

SCAN ME

aka.ms/CopilotStudioImplementationGuide








Q & A

https://forms.cloud.microsoft/r/GE3UZfC0AT

Thank You

Building Business Agents with Microsoft Copilot Studio